A contextualised protein language model reveals the functional syntax of bacterial evolution

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ABSTRACT

Bacteria have evolved a vast diversity of functions and behaviours which are currently incompletely understood and poorly predicted from DNA sequence alone. To understand the syntax of bacterial evolution and discover genome-to-phenotype relationships, we curated over 1.3 million genomes spanning bacterial phylogenetic space, representing each as an ordered sequence of proteins which we used to train a transformer-based, contextualised protein language model, *Bacformer*. By pretraining the model to learn genome-wide evolutionary patterns, *Bacformer* captures the compositional and positional relationships of proteins. We demonstrate the ability of *Bacformer* to: accurately predict protein-protein interactions; uncover operon structure (which we validated experimentally), infer phenotypic traits including antimicrobial resistance, identifying likely causal genes in the process; and design template synthetic genomes with desired properties. Uniquely, *Bacformer* is the first whole-genome model that enables end-to-end genotype-to-phenotype prediction and genome-wide prioritisation of functional loci. Thus, *Bacformer* establishes a genomic foundation model that reveals the evolutionary rules governing bacterial gene organisation, function and phenotype, and opens a route to systematic, whole-genome engineering.

INTRODUCTION

Bacteria are arguably the most successful organisms on Earth, having evolved over millions of years to inhabit diverse geographically environments [1], biological niches [2, 3], and ecosystems [4]; acquiring a vast metabolic repertoire to exploit diverse substrates to generate energy and create biomass [5], and developing complex behavioural traits such as virulence [6], immune evasion [7], and antibiotic resistance [8].

Bacterial genomes are known to encode these adaptations in a highly structured manner. For example: functionally related genes are usually found contiguously in operons [9]; transcription factors mostly regulate proximal rather than distant genes [10, 11]; and the genomic distance from the origin of replication generally predicts gene identity [12, 13]. Thus, the architecture of genomes may provide a framework to create informative representations of bacteria that could be used to understand the basis of evolution and enable generalisable functional predictions across taxa.

Previous large-scale analyses of bacterial genomes have been focused on specific applications such as biosynthetic gene cluster discovery [14, 15] or predicting antimicrobial resistance [16, 17, 18] and have been hampered by biases in public genome collections towards a few pathogenic bacteria [19, 20]. Moreover, existing methods such as DNA and protein language models (LMs) incorporate limited genomic context, ignoring the network of functional interactions encoded across entire bacterial genomes and their structure [21, 22, 23, 24, 25, 26]. In contrast, we sought to develop a foundation model of bacteria that captures the gene structure and function of entire genomes across the full spectrum of bacterial diversity. To do this, we leveraged and curated existing collections of over 1.3M bacterial metagenome-assembled genomes (MAGs) [27, 28], derived from environmental sequencing across diverse habitats to maximise phylogenetic and ecological diversity and encoding almost 3 billion proteins, to create a comprehensive foundation model of bacterial life.

We represent individual bacterial genomes as ordered sequences of proteins and use these as inputs to train a transformer-based, contextualised protein language model (pLM), called *Bacformer*, which learns the functional organisation of bacterial genomes. By pretraining on genome-wide evolutionary patterns, *Bacformer* learns the compositional and positional relationships between proteins and performs strongly on core functional genomic tasks, including protein–protein interaction prediction, and inference of phenotypic traits such as antimicrobial resistance and carbon source utilisation. *Bacformer* can accurately reconstruct genomic organisation including operon structure, which we experimentally validated with long-read RNA sequencing, and can also be used for generative modelling to design template synthetic genomes with desired properties. By modelling the genome as a sequence of protein embeddings, we successfully create a whole genome model, overcoming the limits of context length imposed by DNA-based models, and enabling unified prediction and interpretation of genome-scale phenotypes.

RESULTS

Training a genome-contextualised protein language model

To develop a foundation model that captures evolutionary processes across bacteria, we first curated a broad taxonomic corpus of over 1.3 million metagenome-assembled genomes (MAGs) from over 25,000 species, spanning almost 1,500 orders and more than 70 biomes (***Figure 1A***).

Every individual assembled genome was then modelled as an ordered sequence of proteins, each of which is represented using existing protein language model (pLM) embeddings (from the ESM family [21, 29]) which capture evolutionary and functional properties of proteins [22, 21, 30, 29], supplemented with rotary positional embeddings [31] and special tokens marking contigs; an arrangement that preserves the genomic ordering of proteins (***Figure 1B***). *Bacformer* then utilises these enhanced embeddings of the entire proteome of an organism as inputs to a 12 layer (*Bacformer*) or a larger 30 layer (*Bacformer-L*) transformer model [32], treating each protein sequence as a single token and processing the complete genome in a single pass (see *Supplementary Methods*).

Unlike DNA or protein-based language models (LMs) that rely on pre-defined and constrained token dictionaries [33, 25, 23, 21, 22], *Bacformer* accommodates the vast diversity of possible proteins by clustering them within embedding space into protein families, and using these labels as a discrete vocabulary for the autoregressive pretraining task, where the model learns to predict the family of masked proteins based on the context of other proteins in the genome (***Figure 1B***). To ensure that our protein family clusters reflect genuine evolutionary relationships, we benchmarked them against established orthology databases (*eggNOG* [34]) and clustering methods (*MMseqs2* [35]), obtaining normalised mutual information (NMI) scores of 74% and 88% respectively, indicating high agreement and allowing us to leverage the protein family clusters to learn the contextual interactions across proteomes during pretraining. *Bacformer* is designed to take as input the entire proteome natively supporting fine-tuning for whole genome tasks. Existing DNA and protein-based LMs only can model full genomes by splitting them into segments which requires orders of magnitude more GPU memory for fine-tuning, making this approach intractable to most users (***Figure 1C***).

We observed that the pretraining accuracy and perplexity of *Bacformer* increased progressively during pretraining, improving with dataset and model size, demonstrating robust scaling laws seen with other transformer architectures [36, 37] (***Figure S1***). *Bacformer* generalised across species, to previously unseen genera, and across a range of genome completeness (***Figure S1***; ***Figure S2***). We further refined *Bacformer* by extending pretraining to include 48,932 complete genomes [38] (which we expect to provide important additional contextual information). Having access to complete genome boosted both zero-shot (i.e. no further training) and finetuned performance, highlighting the value of an uninterrupted genomic context for the model (***Figure S2***).

Genome embeddings uncover evolutionary relationships across bacteria

We next explored how the unified representation of bacterial genomes learned by *Bacformer* might provide insights into strain-level diversity and reveal sub-populations with distinct functional or evolutionary characteristics.

Existing clustering methods for bacteria operate only on closely related strains or rely on a limited set of marker genes and fail to consider gene content [39, 40, 41, 42, 43]. In contrast, *Bacformer* represents individual genomes across species in a shared embedding space (by averaging their contextual protein embeddings across the genome), enabling more robust strain-level comparisons based on the organisation of proteins within each isolate, and learning evolutionary relationships in an unsupervised way.

When applied to different sets of species, genera, and families (see *Supplementary Methods*), *Bacformer* accurately clustered genomes according to their taxonomic labels (based on Adjusted Rand Index (ARI), Adjusted Silhouette Width (ASW), and NMI), considerably outperforming non-contextual pLM embeddings (*ESM-2* for *Bacformer* and *ESM-C* for *Bacformer-L*) as well as other genomic LMs, with *Bacformer-L* achieving a score of over 0.98 ± 0.006 and 0.97 ± 0.017 on NMI and ARI respectively (***Figure 1D,E***; ***Figure S3,S4***). Moreover, even in this fully unsupervised manner, *Bacformer* learned phylogenetic relationships. For example, for the *Bacteroidales* order, *Bacformer* embeddings produced average cosine similarities among species belonging to the same genus or to the same family of 0.96 ± 0.02 and 0.93 ± 0.05, respectively (***Figure 1F***).

To evaluate whether *Bacformer* can resolve fine-scale diversity within a single species and identify meaningful evolutionary differences, we examined the *Escherichia coli* sample collection from the *MGnify* dataset [27]. We extracted *Bacformer* embeddings for each MAG and applied *Leiden* clustering [44], which clearly identified six distinct subpopulations (***Figure 1G***) that were not well delineated when we used *ESM-2* embeddings, and were not driven by artifacts, such as genome sample or study origin (***Figure S5***). Using pre-computed graphical pangenome analysis (*Panaroo* [45]), we identified several core and accessory genes that were significantly cluster-specific (adjusted $p < 0.05$, *Fisher's* exact test) including: *cbtA* (in cluster 2) that regulates cell division and elongation [46], *stxA* (in cluster 5) that produces *Shiga* toxin responsible for life-threatening complications [47, 48], and *fldP* (in cluster 3) that shields cells from oxidative stress and aids survival in low-oxygen environments [49, 50] (***Figure S5***).

Bacformer embeddings can also be used to rapidly search for related genomes and infer evolutionary context, ecological niche and bacterial behaviours. We linked the 1.3M *Bacformer* genome embeddings to associated biological and clinical metadata to allow users to input an unannotated genome and retrieve similar genomes and associated phenotypes from our database. We validated the effectiveness of this method by applying a k-nearest neighbour (kNN) classifier to a held-out subset of bacterial genomes, demonstrating highly accurate retrieval across taxonomic ranks (***Figure 1H***), with the method achieving almost 95% balanced accuracy across diverse genera.

Thus, *Bacformer*'s learned representations of complete contextualised genomes can reveal important evolutionary signals at both broad phylogenetic and localised strain-specific scales, and can be used to infer bacterial properties of unseen genomes.

Bacformer reveals functional organisation of proteins

Since *Bacformer* was trained to predict proteins from their genomic context, we anticipated that it would learn the evolutionary relationships between proteins and hence their organisation into functional networks.

To explore this potential emergent property, we first examined the model attention weights across thousands of genes in diverse genomes and found that *Bacformer* focuses on neighbouring genes. While early layers distribute attention evenly, middle layers concentrate attention strongly on closest neighbouring genes, whereas later layers spread attention further along the chromosome (***Figure S6***). *Bacformer* strongly updates embeddings of adjacent proteins (assessed by their cosine similarity), highlighting the importance of rotary positional embeddings in capturing gene order (***Figure S6***), and indicating that it has learnt evolutionarily-conserved arrangements of contiguous proteins.

We therefore tested whether *Bacformer* could predict the presence of operons (physically adjacent co-transcribed genes that represent the fundamental regulatory units of bacterial genomics [9]). Using Operon DB [51] as ground truth, *Bacformer* and *Bacformer-L* accurately predicted operons across distinct strains, out-performing non-contextual *ESM-2* and *ESM-C*, as well as other pLM methods (***Figure 2A***; ***Figure S7***).

Since current operon annotations are incomplete and potentially inaccurate, we performed long-read RNA sequencing of strains from five bacterial species (*Mycobacterium tuberculosis*, *Mycobacterium abscessus*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*, and *Escherichia coli*), and obtained accurate zero-shot *Bacformer* predictions (AUC $0.76 - 0.92$) when evaluated against experimentally validated operons (***Figure 2B***; ***Figure S7***), demonstrating how *Bacformer* captures genome organisation without additional training.

We next evaluated whether *Bacformer* also implicitly learns genome-wide protein-protein interactions and hence would be able to predict functional protein networks. We computed pairwise cosine similarities in protein embeddings for a diverse test set of $2,088$ bacterial genomes across $1,573$ species and employed these in a zero-shot classification task, using different types of interaction score (combined, co-occurrence, neighbourhood, co-expression) from the *STRING* database [52] as binary labels. Unlike traditional co-occurrence or phylogenetic profiling approaches, our strategy avoids multiple-sequence alignments and ad hoc distance cut-offs, yielding a single scalable similarity measure across proteins and species. *Bacformer* considerably outperformed its non-contextual protein language model base (*ESM-2*) across all tasks (***Figure 2C***; ***Figure S8***) and was further significantly improved by finetuning (p-value < 0.01, t-test; see *Supplementary Methods*,

Figure S8). During benchmarking, we saw *Bacformer* and *Bacformer-L* significantly outperform other deep-learning based methods on held-out strains (***Figure 2C***) with *Bacformer-L* improving the zero-shot AUROC by 14 percentage points compared to its base pLM *ESM-C*, demonstrating the importance of incorporating whole-genome context.

When applied to *P. aeruginosa* (PAO1), the finetuned model uncovered previously known, as well as novel, high-confidence protein-protein interactions (***Figure 2D***). We then used *AlphaFold3* [53] to validate the novel physical interactions by predicting the structure of protein complexes (***Figure 2E***). Protein complexes predicted by *Bacformer* had significantly higher *ipTM* scores (a metric that evaluates the accuracy of predicted protein-protein interfaces [54]) than for complexes based on random pairs ($p < 0.003$; Welch's t-test; ***Figure S9***), indicating direct physical binding.

Bacformer was also able to identify broader regulatory networks using its embedding similarity matrix, even without direct input of transcriptional information. For example, in *P. aeruginosa*, *Bacformer* was able to predict clusters of co-expressed genes more accurately than *ESM-2* (***Figure S9***) and identify networks mediating canonical quorum-sensing [55], biofilm formation [56], and motility [57] (***Figure 2F***; ***Figure S9***), showcasing how *Bacformer* embeddings can reveal co-evolved, co-regulated, and interacting protein families.

Predicting protein essentiality and functions

To explore how kingdom-wide evolutionary signals and their genome-wide context learnt by *Bacformer* might reveal protein characteristics, we first examined its ability to predict protein essentiality. To create a ground truth dataset, we compiled genomes from 51 bacterial genomes across 37 species (from the *DEG* database [58]), providing a total of 169,408 genes labelled as essential or non-essential. We then examined the performance of *Bacformer* and *Bacformer-L* in predicting protein essentiality, finding that they outperformed their base models and achieved AUROC on par with the significantly larger *gLM2* model and better than other genomic LMs (***Figure S10***; see *Supplementary Methods*). A linear model trained on top of the *Bacformer* embeddings was able to clearly separate essential from non-essential genes based on predicted probability distributions, achieving 0.81 ± 0.07 median AUROC across diverse genomes, and was able to identify many genes not previously annotated as essential (which we scored as false positives) but with orthologs in other species that are known to be essential, such as ribosomal proteins [59, 60] and tRNA synthetases [61, 62] (***Tables S1*** & ***S2***), indicating that our model issued biologically plausible predictions.

Importantly, we found that *Bacformer* was still able to issue confident predictions on gene essentiality even when we masked the query protein itself (0.64 median AUROC across held-out genomes, ***Figure S10***), indicating how important the learned contextual information is for model predictions.

We then evaluated the ability of *Bacformer* and *Bacformer-L* to predict protein function. We leveraged KEGG [63] to extract protein function labels across diverse genomes (see *Supplementary Methods*) resulting in a dataset of 1,126,510 proteins across 300 diverse genomes and 9,059 unique protein

function labels (here, KEGG KO orthology groups). Using the dataset, we trained a linear layer on top of the protein embeddings to predict the function of each protein. *Bacformer* and *Bacformer-L* achieved balanced accuracy of over 94% across held-out test strains, increasing to over 98.5% after finetuning the entire model (***Figure S11***). We were able to annotate hypothetical as well as other unannotated proteins, allowing us to confidently annotate 98,821 proteins across 60 held-out strains, demonstrating how the *Bacformer* models can be used for protein function inference (***Figure S11***).

Genome-scale inference of bacterial phenotypes and ecological niche

By modelling entire bacterial genomes, *Bacformer* links genome-wide organisation to emergent bacterial phenotypes and ecological niches. We found that *Bacformer* genome embeddings intrinsically reflect phenotypic traits. For example, isolate embeddings clustered by Gram stain (***Figure S12***) and by host environment, even when comparing closely related species, for example within the *Bifidobacterium* genus (***Figure 3A***). We therefore assessed whether *Bacformer* even without finetuning could predict bacterial phenotypes such as metabolism, carbon source utilisation or oxygen requirement.

We compiled a dataset of 24,462 genomes from 15,477 species that had been annotated with 139 distinct phenotypic traits [64, 65, 66] and trained separate classifiers to predict each phenotype from *Bacformer* genome-level embeddings (***Figure 3B***) computed by averaging all of the contextual protein embeddings present in the genome. *Bacformer* and *Bacformer-L* accurately predicted phenotypes across a range of diverse traits including carbon source utilisation, optimal growth temperature, motility, and toxin production, outperforming other methods (***Figure 3C***). We then employed those phenotype-specific models with high predictive performance (balanced accuracy > 85%) to annotate the entire 1.3M genomes within the MAG corpus across diverse phenotypes spanning metabolic, ecological, and physiological characteristics (***Figure 3D***; *see Supplementary Methods*). We provide these models, annotations, and accompanying metadata, which represents the first large-scale resource for kingdom-wide phenotype inference of bacteria, for researchers to explore, filter, or extend for their own studies.

Given the depth of phenotypic information that *Bacformer* can predict for a large phylogenetic spread of bacteria, we wondered whether we could use *Bacformer* to begin to understand the functional nature of polymicrobial communities. As an exemplar, we annotated a single human gut microbiome sample containing 5,811 MAGs with all 139 predicted phenotypes which we used to compute a UMAP plot (labelling by both bacterial family and predicted lactose utilization (***Figure 3E***), and show the phenotypic variability within families and the potentially complex functional interactions *Bacformer* might uncover between species.

The accuracy of *Bacformer* in predicting phenotypes motivated us to further leverage its contextual genomic representations to discover the genes that are likely causally associated with specific traits by employing a gradient-based attribution method [67] that assigns importance scores to genes in a genome. Uniquely, *Bacformer* enables finetuning using a whole-proteome context as input,

allowing us to identify phenotype-specific genes across diverse genomes.

We first examined the ability of this method to identify genes that were identified as related to sporulation, motility, and anaerobic growth (three phenotypes with some pre-existing ground truth knowledge of responsible genes [68, 69, 70, 63]), and found that phenotype-associated genes were assigned higher importance scores than other genes (*Mann-Whitney* test, $p < 10^{-10}$; ***Figure S12***).

We then examined in detail the sporulation-associated genes identified by *Bacformer* and found that: the canonical sporulation genes, *spoIIIAG* and *spoIIIAF* [71, 72], had been ascribed the highest importance scores; and that clustering the top 100 sporulation-associated genes revealed functionally distinct groups including known regulatory factors (*sigG*, *spoIIAB* [73, 74]) and structural genes (*spoVE*, *spoVB* [75, 76]), as well as previously uncharacterised, but plausible, candidates, such as *katB*, *galE*, and *timB* (***Figure 3F***).

We next tested whether we could use *Bacformer* to identify niche-specific genes. Within *Lactobacillus johnsonii*, a gut bacteria found in several animal species [77, 78], *Bacformer* was able to identify known genes involved in host-specific adaptation, such as *lacA* and *lacR* (human, [77]), *cas9* genes in livestock [79, 80], type III RM gene (*mmr*) (pig [79]), and *banIM* (chicken, [81]), as well as a credible set of additional novel genes (***Figure 4F***; Table S3). Finally, applying a similar approach to the *Prevotella*, a genus for which gene annotations and experimental validations are sparse, *Bacformer* was able to identify novel protein families specifically associated with gastrointestinal environments in different animal species (***Figure 3H***).

Thus, by combining large-scale phenotype prediction with high-resolution gene-level attribution, *Bacformer* can not only assign specific traits across bacteria but can also discover potential causally-associated genes.

Predicting antimicrobial resistance (AMR) and AMR-associated genes

We next evaluated whether *Bacformer* could predict antimicrobial resistance (AMR). Using an existing dataset of isolate genome sequences from 38 pathogenic bacterial species and their response to 54 different antibiotics (NCBI AST Browser [82]), we trained a separate linear classifier and regression model for each antibiotic using the genome-level embeddings of each isolate (see *Supplementary Methods*). *Bacformer* and *Bacformer-L* markedly outperformed other DNA and protein language models on both binary classification and regression tasks (***Figure 3I***).

In contrast to other genomic language models which only consider short context windows [25, 33, 21, 23, 26, 24], *Bacformer* is the only method specifically designed to model whole genomes and consequently can be readily finetuned for organism-level phenotypic prediction tasks. To illustrate this property, we finetuned *Bacformer* to predict resistance for three distinct combinations of bacterial species and class of antibiotics, achieving near perfect performance (measured by the area under ROC (AUROC) and Precision-Recall (AUPRC) curves; ***Figure 3J***), even with training sets of less than 800 isolates. Through in silico ablation studies, we could show that *Bacformer* identified both

known and novel AMR-associated genes, supporting its use as a biological discovery tool (***Figure 3J***; ***Table S4-S6***).

Generating de novo bacterial genomes

Since *Bacformer* has learnt genomic representations of bacteria that encompass evolutionary relationships, reflect protein functional organisation, and allow phenotype inference, we next looked to evaluate its potential for synthesising bacterial genomes *de novo*.

We first evaluated *Bacformer's* generative capabilities by analysing its performance on unseen genomes. *Bacformer* achieved low perplexity (a measure of uncertainty when predicting the next token) across diverse genomes (***Figure 4A***; ***Figure S13***), suggesting it captures the genomic architecture of strains not encountered in the training set. We then prompted *Bacformer* with a short sequence of initial proteins from *Escherichia coli* and *Staphylococcus aureus* and asked it to generate a complete genome (see *Supplementary Methods*). We showed that the resulting ordered sequence of protein families within synthesised genomes was comparable in distribution and ontological diversity to those of true genomes (***Figure 4A,B***; ***Figure S13***), demonstrating *Bacformer's* capacity to create genomes that span the full spectrum of protein functions. To ensure that the model did not simply recapitulate genomes seen during training, we generated multiple genomes from identical protein prompts while varying the temperature (a scaling factor in autoregressive sampling that controls randomness) and observed that *Bacformer* generated a variety of distinct genomes, none of which were replicas of any within the training corpus (***Figure S13***).

We then tested whether *Bacformer* could accurately retrieve the full genome from an incomplete set of proteins, a valuable capability particularly for partial genomes typically recovered from metagenomic sequencing approaches. We therefore artificially removed a percentage of proteins from a collection of held-out complete genomes and computed their genome embeddings compared to the complete genomes, calculating the mean reciprocal rank (MRR). *Bacformer* was able to extrapolate the full genomic context even when only 50% of proteins were present, demonstrating the ability to confidently infer whole proteome from sparse protein sets (***Figure 4C***; ***Figure S14***).

Since *Bacformer* was trained to predict protein families rather than specific proteins, we augmented its training with a contrastive loss, encouraging the model's contextual embeddings to align closely with actual protein embeddings present in our database (see *Supplementary Methods*), thereby enabling retrieval of specific protein sequences from generated contextual protein embeddings (***Figure 4D***). To align the protein sequence embedding with the generated contextual embeddings, we finetuned *Bacformer* on a non-causal protein prediction task (where 15% of proteins were masked), combining the masked objective with a contrastive loss. Resulting embeddings were significantly more similar to their true counterparts than to other proteins (p-value $< 10^{-10}$; ***Figure S14***), and demonstrated the capability to retrieve the underlying protein sequence from a contextualised protein representation (***Figure 4E***).

Finally, we explored how *Bacformer* could be used to generate entire bacterial genomes endowed with specific properties, a core ambition of synthetic biology [83, 84, 85]. Leveraging our phenotypic trait analysis (***Figure 3***), we introduced a property token at the start of each protein-family sequence (***Figure 4F***), and then finetuned *Bacformer* to generate protein-family sequences conditional on a particular phenotype, selecting two different traits (oxygen requirement and optimal growth temperature) to evaluate its generative capacity.

We then used each phenotype-specific model to generate full genomes from an initial set of protein families drawn from held-out species and observed functional enrichment of proteins with biological roles linked to the specified phenotype (***Figure 4G***). For example: conditioning on aerobic growth produced significant positive enrichment of urate oxidase (*pucL*) and cytochrome c family genes (*CYC*), both of which catalyse key oxidative steps in purine catabolism and electron transport during aerobic respiration [86, 87], and significant depletion of proteins that promote anaerobic metabolism (such as the nitrate-responsive regulator *narL* and formate-hydrogen lyase component hycA [88, 89]). Conditioning on psychrophilic growth yielded significant enrichment of cold-active xylanases (*xynB*) and β-galactosidases (*bgaB*), enzymes commonly isolated from polar and deep-sea bacteria where they facilitate polysaccharide degradation at low temperature [90, 91], and significant depletion of genes (such as steroid Delta-isomerase (*ksi*) [92]). These findings indicate the potential utility of *Bacformer* for rational genome design.

Discussion

Bacformer, a genomic foundation model trained on millions of diverse bacterial strains, offers a unified representation of genomes, modelled as an ordered sequence of interacting proteins, and generalises across species to incorporate the constraints and relationships that shape proteomes during evolution.

During pretraining, *Bacformer* generates highly informative whole-genome embeddings that identify biologically important clusters of bacteria and their associated gene content (providing important biological and/or clinical insights) and can be used to rapidly annotate unknown genomes by relating them to existing isolates with known phenotypes. Importantly *Bacformer* exhibits emergent properties by learning both protein-protein interactions and protein functional organisation, enabling accurate prediction of operons, protein annotation, and acquiring the syntactic rubric underpinning bacterial evolution and functional diversification, thereby driving biological discovery.

Leveraging these emergent properties, we have trained *Bacformer* to predict bacterial phenotypes, including antimicrobial resistance, and identify the putative causal gene modules responsible, thereby revealing important biological processes, permitting phenotype inference within or across species at microbiome and ecological scale, and potentially enabling state prediction following perturbations.

Bacformer can also be used for generative tasks, inferring complete genomes from an incomplete

set of proteins and creating a road map for de novo genome synthesis with desired properties; with clear applications to synthetic biology and bioengineering.

In summary, *Bacformer* provides both important insights into the evolutionary language of bacterial genomic organisation, function and phenotype, as well as a generative approach to whole genome design.

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Author Contributions

M.W., A.W. and R.A.F. conceived the project. R.A.F., A.W., M.B. and J.L. supervised the project. M.W. designed the model architecture. M.W., A.W. and M.N. curated and preprocessed pretraining datasets. M.W., A.W., R.V.T., D.C.A. curated and preprocessed datasets for finetuning and downstream tasks evaluation. M.W. implemented and models and infrastructure for pretraining.

M.W. evaluated the pretraining model and conducted the finetuning. M.W. D.C.A., A.W. conducted experiments for downstream tasks analysis. D.A. prepared the samples for long-read RNA sequencing. A.W., M.W. and D.A. preprocessed and analysed the long-read RNA sequencing results. R.V.T. conducted experiments on phenotypic traits prediction. D.C.A. conducted experiments on anti-microbial resistance prediction. A.D. contributed to protein-protein interaction prediction analysis. S.H. and J.L. contributed to the modeling infrastructure. M.W., R.A.F., A.W., M.B., S.H. and J.L. contributed to study design and analysis. M.W. implemented the public codebase. R.A.F., M.W. and A.W. wrote the first draft of the manuscript. All authors wrote the final draft of the manuscript.

Competing Interests

Authors declare no competing interests.

Code and data availability

Code and models related to this study, together with tutorials are publicly available at https://github.com/macwiatrak/Bacformer under a permissive *Apache 2.0* License. The code for benchmarking and leveraging different genomic LMs is available at https://github.com/macwiatrak/BacBench.

The genomic sequences and assemblies are all publicly available and downloaded from *MGnify* [27], *SPIRE* [28] and NCBI RefSeq [38]. The long-read RNA sequencing operon annotations are available at https://huggingface.co/datasets/macwiatrak/operon-identification-long-read-rna-sequencing. Downstream tasks datasets used for evaluation and benchmarking are available at https://huggingface.co/datasets/macwiatrak/datasets

Additional details on the datasets used in this study and processing are provided in Methods.

Figure Legends

Figure 1. Bacformer, a bacterial foundation model, represents bacterial genomes as a sequence of contextualised protein embeddings. (A) To pretrain *Bacformer*, we constructed a metagenome-assembled genomes (MAG) corpus, a dataset of more than 1.3 million genomes, 3 billion proteins, and 225 million contigs collected across 25,000 species and 70 distinct environments. For visualisation, the taxonomy tree represents a set of the 200 orders with the highest number of genomes in the MAG corpus, annotated by number of genomes (*log*; dark blue outer circle) and distribution across biospheres. **(B)** *Bacformer* model architecture and pretraining scheme. *Bacformer* takes as input the set of protein sequences present in a bacterial genome, ordered by genome location. Protein sequences are then embedded with a pretrained protein language model [21, 29] and combined with rotary positional embeddings [31] as well as learnable contig representations. The resulting protein representations are then used as input to the transformer, which computes contextualised protein embeddings using other proteins present in the genome. During pretraining, the model predicts a protein family of the masked protein based on the genomic

context. The discrete vocabulary of protein family labels necessary for pretraining was established by embedding the non-contextual protein sequences with a protein language model and performing unsupervised clustering. **(C)** GPU training memory required to model 3 representative genomes using different genomic LMs. The required memory was calculated by generating the cost of forward & backward pass with batch size of 1 on distinct genomes. For all models we used Flash attention and `bfloat16` precision on NVIDIA A100 GPU (see *Supplementary Methods*). **(D)** *Bacformer* accurately clusters MAGs across diverse genera, in an unsupervised manner. UMAP plot showing strain clustering using *Bacformer* embeddings of assemblies from the 10 most frequent genera in the MAG corpus, amounting to 314,148 unique strains (clustering metrics: Adjusted Rand Index (ARI), Normalised Mutual Information (NMI); all metrics vary from 0 to 1, higher is better). **(E)** Performance of *Bacformer* and *Bacformer-L* compared to their base pLMs, *ESM-2* and *ESM-C* respectively, on a species strain clustering benchmark task consisting of 6,071 MAGs across 25 species. Metrics vary from 0 to 1, higher is better. Error bars denote 95% confidence intervals estimated from 10 independent bootstrap resamples, each using 80% of the data. **(F)** *Bacformer* representation of MAGs maintains evolutionary relationships. Cosine similarity heatmap for 100 most common species of *Bacteroidales* showing higher similarity between species in the same family and genus. **(G)** *Bacformer* reveals subpopulations within a single species. UMAP plot of *Escherichia* coli MAGs from the *MGnify* dataset [27] using genome embeddings from *Bacformer* annotated with *Leiden* clustering labels (colours). **(H)** *k*NN classifier results across taxonomic ranks (family, green; genus, light blue; species, dark blue) demonstrate that *Bacformer* accurately identifies similar bacterial isolates. Error bars represent 95% confidence interval across genomes.

Figure 2. *Bacformer* identifies protein-protein interactions across bacterial genomes. (A) Zero-shot operon prediction performance (evaluated by area under the receiver operator characteristic curve; AUROC) across protein-based methods and mixed-modality *gLM2* model on 9 distinct species using annotated operons obtained from OperonDB [51] as ground truth. Line and boxes represent median and interquartile range (IQR) respectively across genomes (median across operons); whiskers extend 1.5 × IQR beyond Q1 and Q3; random classifier AUROC (red dashed line). **(B)** *Bacformer* zero-shot operon predictions (assessed by AUROC) for five bacterial species where we generated experimental ground truth data from long-read RNA sequencing. **(C)** *Bacformer* learns protein-protein interactions in a zero-shot manner. **(i)** Performance across different interaction types from STRING DB [52], outperforming its non-contextual protein embeddings (*ESM-2*, yellow) on held-out genomes ($N = 2,088$). **(ii)** Benchmarking zero-shot performance across different genomic LMs on a smaller randomly sampled set of strains ($N = 54$) from *STRING DB*. The interaction scores were calculated using cosine similarity between protein representations on held-out genomes (see *Supplementary Methods*). Line and boxes represent median and interquartile range (IQR) respectively; whiskers extend 1.5 × IQR beyond Q1 and Q3; random classifier AUROC (red dashed line). **(D)** Proteome-wide interaction scores for *Pseudomons aeruginosa* PAO1 predicted from finetuned *Bacformer* (line thickness indicates prediction strength; only predictions with probability

> 0.5 were visualised). **(E)** Predicted structures obtained from *AlphaFold3* [53] for top scoring protein complexes from *P. aeruginosa* PAO1 predicted by *Bacformer* (*ipTM* score indicates the probability of the two proteins binding together and *pTM* score is a measure of overall predicted structure confidence; range for both 0-1, higher is better). **(F)** Protein-protein interaction network extracted using *Leiden* clustering (resolution=40) [44] on *Bacformer* contextualised protein embeddings. The edge weights with high importance (cosine similarity > 0.8) are highlighted in dark blue.

Figure 3. *Bacformer* enables phenotypic traits prediction and discovery of phenotype-specific genes. (A) *Bacformer* genome embeddings capture environment niche in an unsupervised manner. Cosine similarity heatmap between strains from the *Bifidobacterium* genus showing higher similarity between strains from the same animal niche (chicken gut, pink; pig gut, green; human gut, blue). The dendrogram was created by performing hierarchical clustering on *Bacformer* genome embeddings. **(B)** Phenotype prediction performance of pretrained *Bacformer* (dark blue), without pretraining *Bacformer* (light blue), and non-contextual *ESM-2* (yellow) shown for a range of phenotypes. Line and boxes represent median and interquartile range (IQR) respectively; whiskers extend $1.5 \times$ IQR beyond Q1 and Q3. **(C)** Benchmarking performance across methods on 139 distinct phenotypes across random seeds. For each method-phenotype combination we trained a linear layer on top of the genome embedding splitting the data by genus to ensure generalization across phylogenetically different bacteria (see *Supplementary Methods*). Error bars represent 95% confidence interval across phenotypes and random seeds. **(D)** Example of phenotype inference across bacterial classes. Inner circle: phylogenetic tree of a selection of bacterial genomes (sampled from the MAG corpus), colour-coded by class. Outer circles: a subset of phenotypes (from a total of 139) inferred using phenotype-specific models trained using *Bacformer* genome embeddings. **(E)** UMAP representation of the microbial community from a human gut sample created using the predicted phenotypic traits. Each point is a MAG. MAGs belonging to the species with at least 50 genomes were extracted from a metagenomic sample originating from human gut resulting in $2,132$ unique MAGs. The MAGs were then annotated using *Bacformer* model with 139 phenotypic traits. The UMAP was computed by performing PCA followed by UMAP on the matrix of MAGs by predicted phenotypic traits and coloured by family and lactose utilisation. **(F)** UMAP plot of the 100 genes with the highest importance scores for the sporulation phenotype across genomes (using their embeddings extracted from the last layer of the *Bacformer* model and averaged across genomes) and annotated by *Leiden* clustering [44] label and function (function labels were established by analysing the genes within each Leiden cluster). **(G)** The top 100 genes from *Lactobacillus johsonii* genomes with the highest importance scores for each environmental niche (human gut, brown; chicken gut, dark blue; pig gut, light blue; i.e. isolation source) and plotted as a UMAP (using gene embeddings extracted from the last layer of the *Bacformer* model and averaged across genomes). Some genes with known association to the environmental niche are labelled. Genomes and environmental niche labels provided by the *MGnify* dataset [27]. **(H)** UMAP plot of the top 100 protein families (i.e. clusters) with highest importance scores for each environmental niche (cow rumen, green; human oral, grey;

chicken gut, dark blue; human gut, brown; pig gut, light blue) across *Prevotella* genus genomes using protein family embeddings extracted from the last layer of the *Bacformer* model and averaged across genomes. Genomes and environmental niche labels provided by the *MGnify* dataset [27]. **(I)** Benchmarking AMR prediction performance across methods on distinct drugs across random seeds. For each method-drug combination we trained a linear layer on top of the genome embedding. We evaluated performance in the binary (top and middle; predicting susceptible/resistant labels) and regression setting (bottom; predicting minimum inhibitory concentration; see *Supplementary Methods*). **(J)** Performance of finetuned *Bacformer* model on 3 distinct species-drug combinations on the area under the receiver operating characteristic (AUROC; top) and precision-recall (AUPRC, middle) curves. Error bars represent 95% confidence interval across drugs and random seeds. Volcano plot demonstrating gene element importance discovered by *Bacformer*, annotated by *AMR Finder Plus* [17]. The *x*-axis shows the model attribution (rank-biserial statistic derived from *Mann-Whitney* U rank-sum test with higher values indicating greater positive attribution compared to background). The y-axis displays statistical significance ($-log_{10}$ adjusted *p*-value; one-sided test for element attribution > background). Colours represent the prevalence of each annotated element across all genomes in the species. Details of each gene score are provided in *Tables S4-S6*.

Figure 4. *Bacformer* generates whole bacterial genomes with desired properties. (A) *Bacformer* successfully generates genomes not seen during training. Next protein family prediction (orange) on *Salmonella enterica* (top) and *P. aeruginosa* (bottom) compared to true (blue) protein family distributions. **(B)** *Bacformer* generates genomes encompassing a realistic spectrum of protein functions (predicted, orange; true, blue) for *E. coli* (left) and *S. aureus* (right). We prompted *Bacformer* with a short sequence of initial proteins ($N = 500$) and evaluated the generated genomes against true labels. **(C)** *Bacformer* retrieves true complete genome from incomplete genomes. Mean reciprocal rank (MRR; 0-1, higher is better) of genomes generated from inputs with ablated proteins by zero-shot (light blue) and non-zero-shot finetuned *Bacformer* (dark blue) and *ESM-2* (green). Error bars represent 95% confidence intervals. **(D)** Visualisation showing the process of identifying existing proteins from *Bacformer* generated contextual protein embeddings using next-protein family prediction. **(E)** Retrieval of underlying protein sequences using unmasked (blue) and masked (brown) contextualised protein embeddings across a range of k neighbours. Error bars show standard error. **(F)** *Bacformer* generates template genomes (defined as sequences of protein families) with desired properties. We finetuned *Bacformer* to generate protein family sequences conditional on a property (which we represent as a token appended to the beginning of the sequence). The prompt includes the property token and a set of initial protein families present in the genome which is used by the model to generate the rest of the genome. **(G)** We generated template genomes with different desired phenotypic properties (exemplars shown are oxygen requirement and optimal growth temperature). We then compared sequences produced with and without the property token, calculated the *log*2 fold change for *KEGG* orthology (KO) groups [63], and assessed statistical significance with Fisher's exact test.

Supplementary Materials

Methods

Figures S1 to S14

Captions for Tables S1 to S6

Tables S7 to S15

References

Supplementary Materials for:

A contextualised protein language model reveals the functional syntax of bacterial evolution

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METHODS

Model architecture

The *Bacformer* architecture is designed to mirror how bacterial genomes are structured. It represents complete genomes consisting of different replicons, i.e. the primary chromosome alongside any chromids or plasmids, as well as fragmented isolate genomes and metagenome assembled genomes (MAGs) distributed across multiple contigs, while explicitly capturing the relationships between neighbouring genes.

Bacformer utilises a pretrained protein language model (pLM) to generate embeddings for each protein present in the bacterial genome. These protein-level vectors are then passed to a genome-scale transformer that contextualises every representation across the bacterial genome.

Input embeddings

Every input sample to *Bacformer* is an ordered set of proteins from a bacterial genome (an ordered sequence of genes on one or multiple contigs).

Bacformer represents each bacterial genome as a set of protein sequences in context of its ordered location on the genome. To do so, each protein is treated as a token, and tokens combine information from (1) a protein embedding, (2) a contig (also referred to as chromosome or plasmid) embedding, and (3) a relative position embedding to encode relative distance between proteins, using rotary positional embeddings [1].

To create these, each genome G is first converted into a list of proteins P_i ordered by their coordinates on the bacterial chromosome, plasmid(s) or contigs, so that $G = (P_1, P_2, \ldots, P_N)$ for a genome G with N proteins.

Protein embeddings. A pretrained pLM maps every protein sequence P_i to a matrix of residue embeddings; to get a single vector for a protein we average the residue embeddings,

$$\mathbf{e}_i^{\text{prot}} = \frac{1}{L_i} \sum_{j=1}^{L_i} \mathbf{h}_{i,j} \in \mathbb{R}^D,$$

yielding a fixed-length vector of dimension D. We use protein language models from the *ESM* family, specifically *ESM-2* (12 layers, 35M parameters, $D = 480$) [2] and *ESM-C* (30 layers, 332M parameters, $D = 960$) [3] as the base protein language models for *Bacformer* and *Bacformer Large* respectively.

Contig embeddings. To encode the order of proteins and their location on the genome, we represented contigs using a *contig embedding* $\mathbf{e}_k^{\text{contig}}$ for each of the K contigs that are part of the

bacterial genome. As each contig embedding ($\mathbf{e}_k^{\text{contig}}$) is the same for all proteins on the contig, the model can therefore distinguish, for instance, plasmid-specific gene clusters.

For a protein P_i residing on contig c_i we form a token vector

$$\mathbf{x}_i = \mathbf{e}_i^{\text{prot}} + \mathbf{e}_{c_i}^{\text{contig}}.$$

Special learnable tokens are appended to mark global structure: `[CLS]` at the beginning, `[SEP]` between contigs, and `[END]` at the genome terminus. The ordered sequence

$$\texttt{[CLS]}, \mathbf{x}_1, \ldots, \mathbf{x}_N, \texttt{[END]}$$

preserves native synteny, allowing the transformer to represent functional dependencies such as operons [4].

Rotary positional embeddings. The rotary position embeddings [1] are incorporated in the attention mechanism to encode relative distances between proteins, enabling the model to capture genomic context. This is essential in bacteria, where genes are often organized in operons—compact, co-transcribed clusters whose function and regulation depend on precise ordering and spacing. Rotary embeddings are used instead of sinusoidal schemes to avoid length limitations.

Below we show an example of order in a genomic sequence for complete genomes and MAGs. `chr`, `plasm` and `contig` refer to chromosome, plasmid, and contig, respectively.

Complete genome:
[CLS] chr1_P_1 ... chr1_P_{N_1} [SEP] plasm1_P_{N_1+1} ... plasm1_P_N [END]

MAG:
[CLS] cont1_P_1 ... cont1_P_{N_1} [SEP] cont2_P_{N_1+1} ... cont2_P_{N_2} [SEP] ... contZ_P_N [END]

During pretraining on MAGs, we randomize the order of contigs within every bacteria for every epoch to prevent the model from learning spurious signals from the arbitrary contig ordering produced by the metagenomic assembly.

Finally, each token is processed by a shared linear layer, followed by layer normalization and dropout (with a rate of 0.2). The resulting sequence of embeddings for each bacterial genome is then fed back into the transformer, which updates each protein representation by attending to all other proteins in the genome, thereby capturing their genomic context and relationships.

Transformer block

Each bacterial genome, consisting of N proteins, is fed into a transformer to learn protein embeddings conditioned on all other proteins in the genome, yielding contextualised representations. The

rationale is that a genome can be viewed as an ordered set of proteins whose interactions ultimately determine the organism's phenotype.

We employ rotary positional embeddings (RoPE) [1] within the self-attention mechanism and use `FlashAttention` [5] in `PyTorch` for memory-efficient training and inference. The transformer is trained from scratch. For downstream tasks, we use the final hidden state of each protein, while the mean of all token representations provides a fixed-length embedding of the whole genome.

Model sizes (*Bacformer* and *Bacformer Large*)

We trained two versions of *Bacformer*, *Bacformer* and *Bacformer Large (Bacformer-L)* with 26M and 332M parameters in the encoder respectively. The Bacformer (26M) model consists of 12 transformer layers with 8 attention heads and the hidden dimension size of 480. While the Bacfomer Large (332M) has 30 layers, 15 attention heads and hidden dimension size of 960. The architecture of the two is the same, with the difference that in *Bacformer-L* we use SwiGLU activation function [6] instead of GELU [7] and we drop all of the special tokens for the *Bacformer-L*. The choice has been motivated by empirical evaluation.

Throughout the text we refer to *Bacformer* as the 26M parameter model, and *Bacformer Large* as the 332M parameter model.

Pretraining

We sought to learn the relationships between proteins present in the genome, leveraging billions of years of bacterial evolution. All large language models (LMs), including protein LMs and DNA LMs, rely on a discrete vocabulary set for pretraining. In *Bacformer*, each token is a protein, and the universe of bacterial proteins is effectively unbounded. To overcome this, we constructed a finite protein-family vocabulary.

We ran unsupervised clustering on millions of proteins across organisms using k-means clustering [8]. We leveraged *ESM-2* [2] and *ESM-C* [3] protein embeddings to represent the proteins and, after experimenting with various values of k, set it to $50,000$ centroids. Pre-training strongly relies on the unsupervised clustering producing high-quality protein-family classes that reflect evolutionary processes. Therefore, after clustering we performed quality checks leveraging the *ESM-2* protein embedding clustering—evaluating distances to the centroids and the number of proteins per cluster. We removed labels for proteins that exceeded a Euclidean distance threshold of 2.0 to the closest centroid. We leveraged the *CuPy* library [9] to efficiently perform clustering on millions of proteins.

To confirm that these clusters aligned with known evolutionary relationships, we used *eggNOG* 5.0 [10], a database of protein orthology across organisms. Using eggNOG annotations available through *MGnify* [11] on a set of $12,630$ diverse species-representative genomes, we extracted annotations for all proteins with an *eggNOG* label and compared them to the cluster labels from the k-means algorithm. We found strong agreement between the *eggNOG* annotations and our

clusters, with the two achieving a 74% normalised mutual information score (NMI; range 0–1, higher is better), indicating that our k-means clusters capture independently inferred groupings.

Moreover, we compared the *k*-means clustering with *MMseqs2*, a widely used protein-clustering method [12]. We randomly sampled 12.3 million proteins from species-representative genomes, clustered them with *MMseqs2*, and observed 88% NMI concordance with our *k*-means solution.

The protein-family label is predicted using a two-layer multilayer perceptron (MLP) that receives the output of the last hidden layer of the transformer. The MLP is trained from scratch over the course of pre-training. Importantly, we only use the protein-family labels for supervision during pre-training; the input protein embeddings are constructed from amino-acid sequences, not from family labels. This design allows the model to tackle tasks that require amino-acid-level sensitivity. We also experimented with using protein-family labels as token IDs learned from scratch, but amino-acid-level input yielded better performance on downstream tasks.

Training objective. We trained two versions of *Bacformer* with different self-supervised objectives: (1) an autoregressive next-protein-family prediction (generative) objective and (2) a masked-prediction objective. In practice these approaches confer complementary strengths on downstream tasks. Each model contains 12 transformer layers followed by a classification head that predicts the protein-family label from the final layer's output.

The masking procedure resembles the standard BERT-style scheme [13], adapted to our setting: (*i*) 15 % of the proteins are selected for masking. (*ii*) In 87.5 % of those cases the selected tokens are replaced by `[MASK]`. (*iii*) In the remaining 12.5 % the selected tokens are left unchanged.

For *Bacformer-L*, we only trained the model with a masked-prediction objective.

Pretraining details. We first pretrained both models on a corpus of almost ~1.3M metagenome-assembled genomes (MAGs). Training was performed on a mixture of NVIDIA A100 and Intel Ponte Vecchio (PVC) GPUs for up to 1.2 M steps; running on four A100s required approximately two weeks. The effective batch size was 32. We used a linear learning-rate schedule with a 10 % warm-up, ramping to a peak of 0.00015 per GPU; the rate was then scaled by the square root of the total number of GPUs. Optimisation employed Adam [14] with weight decay 0.01 and $\beta = (0.9, 0.999)$. All tensors used the `bfloat16` format. For *Bacformer-L*, we kept the same pretraining hyperparameters, except we used a learning rate of 0.0002 per GPU and performed the training on the NVIDIA H100 GPUs. Models were implemented in `PyTorch` [15] and trained with the `HuggingFace Transformers` library [16].

After this stage we continued pretraining on a held-out set of 48,932 complete genomes. We separated the two stages because MAGs usually consist of multiple contigs whose order is unknown,

whereas complete genomes comprise an uninterrupted chromosome (or set of replicons) with a well-defined gene order, which can be fed to the model as a single contiguous sequence. For every genome we capped the maximum number of proteins at 6,000, a value that exceeds the protein count of more than 99% of genomes in our corpus.

All pretrained models are available for zero-shot use or fine-tuning at https://huggingface.co/collections/macwiatrak/bacformer. Code for pretraining is provided in our GitHub repository: https://github.com/macwiatrak/Bacformer.

Pretraining performance analysis. During training, we monitored the validation loss and selected the checkpoint with the lowest value. We then evaluated pretraining performance by running the model on the validation set at regular intervals, recording accuracy for the masked-prediction objective and perplexity for the autoregressive objective. In both cases performance improved steadily with additional steps, demonstrating that the models generalise to unseen proteins (i.e. zero-shot; ***Figure S1***).

To probe this further we conducted an ablation study across taxonomic ranks in the zero-shot setting. *Bacformer* retains high accuracy on unseen species, but performance declines for genomes that belong to entirely unseen genera, families or orders (***Figure S1***) as the greater evolutionary distance of these taxa reduces overlap in k-mer composition, gene content and synteny between the training set and the target genomes.

We next examined the effect of genome completeness. MAGs are often fragmented and may lack sizeable genomic segments; therefore, the model must be robust to varying completeness levels. pretraining results show that *Bacformer* benefits from more complete assemblies. However, it remains relatively robust until completeness drops below 75 % (***Figure S1***). The same trend appears when comparing MAGs with complete genomes (***Figure S2***): access to the full sequence and an intact chromosome without contig breaks markedly boosts accuracy.

Finally, we investigated scaling. A variant trained on only 25% of the data and with 6 (rather than 12) layers performed noticeably worse, indicating that *Bacformer* benefits from both additional training data and model capacity; further scaling is therefore likely to yield significant gains (***Figure S2***). The benefits of scaling are also visible by the improved performance of *Bacformer-L* over *Bacformer* on the downstream tasks.

Data

We collated a diverse set of bacterial genomes spanning the tree of life and a wide range of environments. Our guiding principle was to maximise phylogenetic and ecological diversity so that the pretrained model would be broadly applicable to downstream tasks in bacterial genomics.

MAG corpus

For pretraining we assembled the MAG corpus from two complementary resources: *MGnify* v2.0 [11] and *SPIRE* [17]. *MGnify* provides seven biome-specific genome catalogues that together contain 306, 596 non-redundant metagenome-assembled or isolate genomes representing 11, 048 species. *SPIRE* provides 99, 146 consistently processed whole-genome shotgun metagenomic samples drawn from 739 studies worldwide and 1, 008, 689 MAGs.

We downloaded the assemblies containing protein annotations and associated metadata from both resources. Protein order within each contig was preserved so that positional encodings could capture synteny. Lineage labels (species, genus, family) and contextual metadata (sample identifier, biome, etc.) were processed as provided by *MGnify* and *SPIRE*.

The final MAG corpus after quality checks comprises of 1, 314, 344 genomes, totalling 2, 700, 709, 306 proteins, 220, 207, 650 contigs, 24, 663 species and 70 distinct environments.

Complete-genome corpus

Many downstream applications—e.g. comparative genomics, gene-context analysis and operon prediction—benefit from access to complete isolate genomes. Unlike MAGs, which are reconstructed *in silico* from a meta-genome-wide assembly and are often fragmented, incomplete or contaminated, complete genomes are obtained from clonal cultures and typically consist of a single circular chromosome (and any additional chromids or plasmids) with high continuity and accuracy [18]. To capture these characteristics, we downloaded available bacterial assemblies from NCBI RefSeq [19] and GenBank [20]. After removing duplicates we retained 48,932 unique complete genomes, each with its translated proteome and annotation files. The metadata (i.e assembly accessions) were stored alongside the sequences. The NCBI collection is biased towards clinically important pathogens—for example, *Salmonella enterica*, *Staphylococcus aureus* and *Mycobacterium tuberculosis*. The combined corpus, consisting of the MAG and complete-genome corpora therefore encompasses a heterogeneous yet complementary training set that balances ecological breadth with assembly quality, positioning the model to generalize across both environmental and clinical bacterial genomics, despite the current available datasets being biased with broad sections of the global bacterial ecosystem under-represented.

Embedding genomes with a protein language model

Bacformer represents each genome by the average of its protein embeddings computed with a pretrained protein language model (pLM). To avoid the cost of re-embedding every protein at every training epoch, we generated all embeddings once, off-line, and cached the resulting protein embeddings.

We chose the 12-layer, 35M-parameter variant of *ESM-2* [2] for *Bacformer* and 30-layer 332M-parameter variant of *ESM-C* [3] for *Bacformer-L* because it offers a good balance between accuracy and computational efficiency when compared with larger pLMs such as *ESM-2-650*M or *ProtT5-XL*

[2, 21]. Embedding the entire corpus of ~1.3 million genomes required approximately 30 days on a cluster equipped with 16 Intel Ponte Vecchio GPUs, and produced ~5.6 TB of embedding files.

Long-read RNA sequencing for operon identification

To test *Bacformer*'s ability to predict operons, we performed long-read RNA sequencing to annotate operons across five distinct strains, processing three independent biological replicates for each strain.

Bacterial strains and culture conditions. Five bacterial strains were used in this experiment: *Staphylococcus aureus RN450* (*S. aureus* RN450), *Mycobacterium abscessus* ATCC 19977 (*M. ab*), ΔleuD ΔpanCD *Mycobacterium tuberculosis* H37Rv 102J23 (ΔleuD ΔpanCD *M. tb*), *Pseudomonas aeruginosa* PAO1, and *Escherichia coli DH5α*. Each strain was cultured in triplicate under nutrient-rich conditions until mid-exponential phase (OD_{600} = 0.4–0.6) was reached.

RNA isolation. Cells were harvested by centrifugation and processed for total RNA extraction using the MasterPure Complete DNA and RNA Purification Kit (Lucigen) with strain-specific modifications: For *S. aureus* RN450, cell pellets were pre-treated with lysostaphin (Tris buffer, pH 8.0) at 37 °C for 30 min to aid lysis. For the mycobacterial strains (*M. ab* and ΔleuD ΔpanCD *M. tb*), cells were mechanically disrupted by bead beating in lysis buffer, followed by extraction with the standard kit protocol. Isolated RNA was treated twice with TURBO DNase (Invitrogen) to remove residual genomic DNA and purified using RNA Clean & Concentrator columns (Zymo Research). RNA integrity was assessed on an Agilent TapeStation RNA ScreenTape system, and concentrations were measured with a Qubit fluorometer (Invitrogen).

Library preparation and sequencing. For each replicate, 1,000 ng of total RNA underwent rRNA depletion with riboPOOLs (siTOOLs Biotech). The depleted RNA was polyadenylated with poly(A) polymerase (PAP) in the presence of a manganese catalyst, adding 50–90 adenosines per molecule. cDNA libraries were prepared with the Nanopore cDNA-PCR kit, pooled and sequenced on a PromethION device equipped with R10 flow cells (Oxford Nanopore Technologies).

Downstream tasks & analysis

Benchmarking

To evaluate *Bacformer* against other methods, we have performed rigorous benchmarking comparing *Bacformer* to other deep learning methods. We selected a diverse set of genomic LMs, including DNA Language Models (DNA LMs), protein language models (pLMs) and mixed-modality genomic LM. These genomic LMs take as input DNA, proteins, or both (*gLM2* [22]) and can therefore generalize across the bacterial tree of life. Moreover, the suite spans modalities (DNA-only, single-protein,

mixed DNA–protein, genome level protein), objectives (masked vs. autoregressive), and training corpora (bacteria-specific vs. cross-kingdom), enabling a controlled comparison of how context length, modality, and pretraining data shape bacterial genome embeddings. The details of the benchmarked methods can be found in ***Table S7***.

Below we outline how we represent and embed bacterial genomes using DNA LMs, pLMs, mixed-modality LMs (*gLM2*) and *Bacformer*.

DNA Language Models (DNA LMs). For gene- and system-level tasks, such as protein function and operon identification tasks respectively, we embed the coding sequence together with its upstream promoter. When the input sequence exceeds the model context (length C), we apply a sliding window of length C with stride S, embed each window, and average the resulting window vectors to obtain the final gene representation. At the encoder level, we retain the last-layer hidden states and use mean pooling across tokens to form each window vector. For genome-level tasks such as antimicrobial resistance prediction, we treat the assembled genome analogously: tile the genome into overlapping C-base windows (processing each contig separately if multiple contigs are present), embed each window, first average within contigs when applicable, and then average across contigs to yield a single genome embedding.

For Evo, which is a long-context DNA LM, we include the full flanking sequence from both sides equally when available up to the maximum context length C. However, the gene vector itself is computed by averaging only the hidden states of tokens that fall within the gene's coordinates, which empirically outperforms averaging the entire slice.

Protein Language Models (pLMs). Given a protein of K amino acids, we take the final-layer hidden states and compute a mean over residues to obtain its vector representation. For genome-scale representations, we compute a per-protein vector for every protein encoded in the genome and then average across all proteins to obtain the genome-level embedding; gene- and operon-level tasks restrict the aggregation to only the proteins involved. Throughout, mean pooling over token embeddings performed better than using a special $[CLS]$ token, so we adopt mean pooling consistently.

***gLM2* (mixed-modality LM)**. *gLM2* ingests mixed-modality genomic scaffolds in which coding regions are translated into amino-acid tokens while intergenic regions remain as nucleotide tokens, thereby preserving local genomic context across adjacent genes. We tokenize each scaffold up to the context C (e.g., $4,096$ tokens). Longer scaffolds are handled with the same sliding-window procedure as for DNA LMs; we keep the last-layer hidden states for each window, mean-pool to a window vector, and average across all windows (and across contigs when present) to obtain genome-level embeddings. For tasks which require gene/protein embeddings, we include flanking sequence

to supply regulatory and positional cues, but the gene vector itself is computed by averaging only the hidden states of tokens that fall within the gene's coordinates, which empirically outperforms averaging the entire slice.

Bacformer. *Bacformer* constructs contiguous, genome-aware inputs built from local protein vectors produced by a base pLM. Specifically, we compute per-protein vectors as above, order them by genomic coordinates (chromosome followed by plasmids), and feed the ordered sequence into a genome-level Transformer encoder with rotary positional embeddings. This produces contextualized protein embeddings that encode both the protein sequence and its genomic neighborhood at a whole-genome level. Averaging these contextualized vectors yields a compact genome representation. During pre-training, *Bacformer* effectively learns to predict co-evolving proteins, encouraging the model to capture functional coupling across the genome.

Code for embedding genomes using the benchmarked methods can be found at https://github.com/macwiatrak/BacBench. We include all the details on the context length of each model (C) and stride used (S) in ***Table S8***.

Computational complexity analysis

We profiled peak training GPU memory on three representative bacterial strains: (1) *Mycoplasmoides genitalium*, (2) *Escherichia coli*, and (3) *Streptomyces coelicolor* chosen to span genomes of markedly different lengths. For each model and strain, the code reports the peak memory for a single full-genome batch ($B = 1$) as the sum of model state memory and saved activation memory. The number of trainable parameters is obtained from each model based on the model's architecture using available weights.

We assume mixed precision with bfloat16 for weights and gradients and FP32 for master weights and Adam moments. Concretely, the code allocates 2 bytes/param for weights, 2 bytes/param for gradients, 4 bytes/param for FP32 master weights, and 8 bytes/param for Adam's first/second moments, yielding 16 B/param in total.

We assume FlashAttention [5] for all architectures, so the $T \times T$ attention matrices are never materialised. For a sequence length of T and a Transformer with hidden size D, feed-forward F, and layers L, the saved activations per saved layer are linear in T: residual states $(T \cdot D)$, $Q/K/V$ projections $(3 \times T \cdot D)$, and MLP activations $(T \cdot F)$. Total activation memory is the sum of these terms multiplied by the number of saved layers and by the element byte-width; we store activations in bfloat16 (2 bytes/element). For Evo, which replaces attention with Hyena operators, there are no $Q/K/V$ projections and we only calculate the residual $(T \cdot D)$ and MLP $(T \cdot F)$ terms —thus preserving the same linear-in-T scaling.

Strain clustering

We computed strain embeddings by averaging the contextual embeddings of all proteins within each genome and used these vectors for clustering. Because *Bacformer* takes only the proteins present in a genome as input—without access to any species labels—it operates in a fully unsupervised manner. Using a pretrained model, we extracted strain representations for the 20 most popular species, 10 genera and 7 families. This yielded $199,195$; $314,148$ and $647,437$ strains in the species-, genus- and family-level datasets, respectively.

For each set we performed Leiden clustering [23] (resolution 0.5) with Scanpy [24]. We benchmarked the results against an *ESM-2* baseline, where protein embeddings are non-contextual. Three clustering metrics were computed:
(1) Average Silhouette Width (ASW; range $[-1, 1]$),
(2) Adjusted Rand Index (ARI; range $[0, 1]$), and
(3) Normalised Mutual Information (NMI; range $[0, 1]$).

Across all taxonomic ranks *Bacformer* significantly outperformed *ESM-2*. ASW decreased at coarser taxonomic levels of genus and family than on species.

To test whether *Bacformer* still preserves intra-family similarity, we computed average species embeddings for the most common bacterial order in *MGnify* [11], *Bacteroidales* ($N = 64,713$), and calculated cosine similarities between the 100 most prevalent species. Species embeddings were obtained by averaging the embeddings of all strains belonging to that species. *Bacformer* retained high similarity within taxonomic groups, achieving an average cosine similarity of 0.96 ($SD = 0.02$) for species in the same genus and 0.93($SD = 0.05$) for species in the same family, versus 0.81 ($SD = 0.03$) and 0.80 ($SD = 0.03$) for species from different genera and families, respectively. These results show that *Bacformer* can robustly cluster species while maintaining phylogenetic relationships in an unsupervised setting. All strain-clustering experiments used the model pretrained on MAGs with the causal (autoregressive) objective.

Revealing sub-populations in *Escherichia coli*. To evaluate whether *Bacformer* can discover sub-populations within a single species, we clustered embeddings of *E. coli* strains. Average genome embeddings for $7,585$ *E. coli* strains from *MGnify* were clustered with *Leiden*, yielding six well-defined groups (***Figure 2G***). Repeating the analysis with *ESM-2* embeddings failed to resolve distinct clusters (***Figure S5***).

To exclude the possibility that the observed structure arose from technical batch effects, we asked whether strains grouped by sample or study identifier (***Figure S5***). The ASW for both metadata labels was negative, indicating no such grouping and confirming that *Bacformer* captures genuine biological structure.

Having identified the sub-populations, we sought the genes specific to each cluster. Using *Panaroo* [25], which provides a binary gene-presence matrix, we performed Fisher's exact tests [26] comparing each cluster against all others. Genes significantly enriched in a cluster ($p < 0.05$) were considered differentially abundant; the top five per cluster are shown in (***Figure S5***). Examples include *cbtA* (cluster 2; inhibits cell division and elongation [27]), *stxA* (cluster 5; Shiga toxin production [28]) and *fldP* (cluster 3; adaptation to low oxygen [29, 30]). This demonstrates that *Bacformer* embeddings can uncover sub-populations with functionally distinct gene complements.

Variation within *E. coli* is often driven by the accessory genome [31]—genes present in only a subset of strains. Defining rare genome as all genes present in $< 5\,\%$ of all strains, we found that 41% of cluster-specific genes were rare. Given that the median genome of *Escherichia coli* contains 112 rare genes versus $3,718$ core genes (those present in$>$ 95% of all strains), this represents a 23-fold enrichment ($p < 10^{-20}$), highlighting the role of the accessory genome in shaping sub-populations and the utility of *Bacformer* for their discovery.

Scalable search for similar bacteria. Leveraging its learned genome representations, We used *Bacformer* to embed all $1,314,344$ genomes in the MAG corpus—together with metadata—into a unified database. Users can submit an unannotated bacterial genome, retrieve its nearest neighbours and obtain contextual metadata such as environmental origin.

We validated this approach with a k-nearest-neighbour (kNN) classifier. Using $k = 10$, we randomly held out 10% of genomes, filtering out species represented by fewer than five genomes. A kNN model trained on the remaining embeddings achieved balanced accuracies of 0.83 ± 0.32, 0.94 ± 0.17 and 0.95 ± 0.16 at the species, genus and family levels, respectively—demonstrating robust performance across taxonomic ranks. This validated functionality is openly available on github (`https://github.com/macwiatrak/Bacformer`), enabling researchers worldwide to analyze unannotated bacterial genomes with unprecedented scale and accuracy.

Benchmarking. We compared *Bacformer* to other protein-based methods and mixed-modality model *gLM2* [22]. We assembled a randomly sampled dataset of $6,071$ MAGs from *MGnify* [11], spanning 25 species across 10 genera and 7 families, selected to provide a balanced phylogenetic spread. For every model, we take the final-layer hidden states and mean-pool across the entire genome to yield one fixed-length embedding per strain.

To compute strain-clustering metrics, we run Leiden clustering over a parameter grid, varying the resolution in [0.1, 0.25, 0.5, 1.0] and the number of neighbours in [5, 10, 15], and evaluating all pairwise combinations. We then retain the combination of resolution and number of neighbours which maximizes the mean of the three metrics described above (ARI, NMI and ASW) at the species-level. Full results are available in the ***Table S9***. We used scanpy [24] to perform the clustering.

Code for strain clustering benchmarking is available at https://github.com/macwiatrak/BacBench/tree/main/bacbench/tasks/strain_clustering and the strain clustering datasets used for benchmarking are available via HuggingFace at https://huggingface.co/datasets/macwiatrak/bacbench-strain-clustering-dna-small (DNA) and https://huggingface.co/datasets/macwiatrak/bacbench-strain-clustering-protein-sequences-small (protein sequences).

Operon identification

Attention-weight analysis. To evaluate how the model utilises gene-neighbourhood information during pretraining, we analysed the attention weights of a model trained with a masked-language objective on complete genomes. Attention scores were collected for 45 diverse organisms present in OperonDB [32].

In *Pseudomonas aeruginosa* PAO1 the model assigns its highest attention to the five closest neighbouring genes (***Figure S6***). Layer-wise inspection revealed distinct patterns: weights in the initial layers are distributed relatively uniformly across neighbouring proteins, mid-layers concentrate on the immediately adjacent genes, and the final layers still focus on neighbours but shift the strongest weights slightly farther away (***Figure S6***). These patterns indicate that successive transformer blocks capture complementary contextual information.

We next examined how *Bacformer* updates neighbouring protein representations. The cosine similarity between adjacent gene embeddings follows an oscillatory pattern reminiscent of a cosine curve (***Figure S6***), demonstrating that the model makes strong use of the rotary positional embeddings that encode genomic position. This observation is consistent with the layer wise attention patterns and underscores the importance of gene neighbourhood.

Zero-shot operon prediction using long-read RNA sequencing. We mapped the reads from each replicate using minimap2 [33] and SAMtools [34] to genes present in the genome assemblies. We then merged the predictions from the triplicates for each strain and used these as our operon annotations. For each sequenced strain, the following genome assemblies and gene annotations from NCBI RefSeq [35] were used:

- `GCF_000013425.1` (*Staphylococcus aureus RN450*),
- `GCF_000195955.2` (ΔleuD ΔpanCD *Mycobacterium tuberculosis H37Rv 102J23*),
- `GCF_000069185.1` (*Mycobacterium abscessus ATCC 19977*),
- `GCF_000006765.1` - (*Pseudomonas aeruginosa PAO1*),
- `GCF_002899475.1` (*Escherichia coli DH5*α).

To predict operons, we posed operon prediction as a binary boundary prediction problem, evaluating whether two contiguous genes form an operon or not. We established a method that utilises: (1) contextualised protein embeddings from *Bacformer*, (2) the distance between the two genes

(intergenic region in base pairs), (3) the strand of each gene. We then constructed a model that takes these features as input and outputs a binary value indicating whether the gene pair forms an operon, based on:

$$s_i = c_i\, e^{-d_i/\lambda}\, I_{\text{strand}}\, I_{\text{gap}},$$

where

$$c_i = \tfrac{1}{2}\big(1 + \cos(\hat{h}_i, \hat{h}_{i+1})\big), \qquad \lambda = 500 \text{ bp},$$

and

$$I_{\text{strand}} = \begin{cases} 1 & \text{if the two genes are on the same strand,} \\ 0 & \text{otherwise,} \end{cases} \qquad I_{\text{gap}} = \begin{cases} 1 & \text{if } d_i \le 500 \text{ bp,} \\ 0 & \text{otherwise.} \end{cases}$$

Here $\hat{h}_i$ and $\hat{h}_{i+1}$ are the ℓ_2-normalised *Bacformer* embeddings of genes i and $i{+}1$, d_i is their intergenic distance, and $s_i \in [0, 1]$ serves as the operon-membership score; a pair is classified as belonging to the same operon when s_i exceeds a threshold. The exponential factor $e^{-d_i/\lambda}$ reduces the score of pairs separated by larger intergenic distances, while the two indicator terms I_{strand} and I_{gap} act as a hard veto, nullifying the score if genes are transcribed in opposite directions or if the gap exceeds 500 bp. This reflects the empirical observation that bacterial operons almost always consist of closely spaced, co-oriented genes.

Benchmarking. We performed benchmarking by utilising *(i)* OperonDB [32], which is a curated database of operons across species, *(ii)* proprietary long-read RNA sequencing results. For both datasets, the evaluation was done in a zero-shot manner.

OperonDB [32] provides partially annotated genomes, therefore, we carefully preprocessed the data, removing non-contiguous operons, genomes with a small number of annotated operons (<8), resulting in 3,335 unique operons across 9 diverse genomes (***Figure S7***). When calculating the operon score for contiguous genes to evaluate models' performance, we restricted ourselves to features that were reliably available for all genes—specifically, those derived from protein sequences—while excluding genomic features such as intergenic distance. To predict operons, we calculated an 'operon score'. The first step in this score was computing the pairwise cosine similarity between the final-layer embeddings of all gene pairs. For a set of colocated genes of size N, the operon score then defined as the average of these pairwise similarity scores across the n genes, providing a single summary value that reflects the overall similarity among proteins encoded within the co-located genes. For evaluation, operons annotated in *OperonDB* were used as positive examples. However, because *OperonDB*'s genome annotations are incomplete, regions without operon annotation cannot be confidently assumed to be true negatives. To address this, negative examples were generated by randomly sampling contiguous gene sets from the same genomes, using a 1:10 ratio of positive (annotated operon) to negative (random gene set) samples. To further

minimize the likelihood of including true, but unannotated, operons among the negatives, this random sampling was repeated five times with different random seeds, and results were averaged across these replicates. We report the performance across strains. We note that the performance varied considerably depending on operon size and species, which likely reflects the differing numbers of verified operons of various lengths and across organisms that are incompletely annotated in *OperonDB*.

To benchmark models on a set of genomes with complete annotations, we used the **long-read RNA sequencing** results generated for this study. To focus the evaluation on the quality of the embeddings, we did not use the intergenic distances and used the cosine similarity between contiguous gene/protein embeddings as well as information on strand to evaluate performance. Specifically, we posed operon prediction as a binary boundary prediction problem, evaluating whether two contiguous genes form an operon or not. Given c_i as the cosine similarity between two contiguous genes and I_{strand} as a binary value indicating whether the two genes are on the same strand, the score for the two genes belonging to the same operon is calculated as $s_i = c_i \times I_{strand}$. We report performance averaged across all the 5 sequenced strains on the area under the receiver operating characteristic (AUROC) and precision-recall (AUPRC) curves.

The preprocessed datasets for operon identification are available at https://huggingface.co/collections/macwiatrak/bacbench. The code used for benchmarking is available at https://github.com/macwiatrak/BacBench/tree/main/bacbench/tasks/operon.

Protein-protein interaction prediction

Protein–protein interaction prediction using *STRING DB*. We downloaded the complete data archive from the *STRING DB* (v12.0) [36] download site (https://string-db.org/cgi/download). Using the species-metadata file, we selected only bacterial organisms and retrieved their protein sequences together with all available protein–protein interaction scores. After running the download scripts, we obtained 10,533 unique strains spanning 6,956 species. *STRING DB* provides several evidence-channel scores; here we used the *combined*, *co-occurrence*, *neighbourhood* and *co-expression* channels.

To binarize the interaction scores we applied a threshold of 0.6 ($\geq$ 0.6 = interaction, $<$ 0.6 = no interaction), resulting in ~10% of all interactions having a positive label. This cut-off was selected empirically by maximising the average AUROC and AUPRC on a validation set. The data were randomly split into training, validation and test sets in a 70/10/20% ratio.

For zero-shot *Bacformer* predictions and evaluation, we computed pairwise cosine-similarity scores between contextualised protein embeddings for every strain in the test set. *Bacformer* embeddings were taken from the last hidden layer of the transformer, pretrained on complete genomes with a masked objective. We benchmarked *Bacformer* predictions for interactions against *ESM-2* [2], whose non-contextual embeddings serve as input to the *Bacformer* transformer (AUROC in ***Figure***

SC; AUPRC in ***Figure S8***). Metrics were calculated per interaction type across genomes.

After confirming that *Bacformer* captures protein–protein interactions in a zero-shot setting, we finetuned the model by stacking a two-layer multilayer perceptron (MLP) on top of the pretrained encoder. The MLP takes as input the mean of the two contextual protein embeddings and is preceded by a dropout layer (frequency=0.2), followed by a GELU activation [7], a normalisation layer ($\epsilon = 1 \times 10^{-12}$) and another dropout layer (frequency=0.2). The resulting vector for each protein pair is passed through a final binary-classification layer, again preceded by dropout 0.2. Training minimises binary cross-entropy loss; we trained for at most 10 epochs with no early stopping, clipped gradients at 2.0, and used Adam [14] with weight decay 0.01, monitoring validation loss across genomes. Hyperparameters were tuned on the validation set. The finetuned model was initialised from the *Bacformer* weights pretrained on complete genomes.

We visualised AUROC and AUPRC (***Figure S8***) curves for representative strains from distinct species in the test set using the predicted interaction scores from the finetuned *Bacformer* model.

We make the *Bacformer*model finetuned for PPI prediction on the *STRING DB* dataset available at https://huggingface.co/macwiatrak/bacformer-for-ppi-prediction. The preprocessed dataset is available at https://huggingface.co/datasets/macwiatrak/bacbench-ppi-stringdb-protein-sequences.

Identifying protein–protein interactions in *Pseudomonas aeruginosa*. To systematically identify potential protein–protein interactions in *Pseudomonas aeruginosa* PAO1, we employed the finetuned *Bacformer* interaction prediction model. To prevent data leakage, the training data excluded strains from the same genus as *P. aeruginosa* (***Figure 2D***). Interaction probabilities were computed for all possible protein pairs, and only those with a predicted probability of at least 0.5 were retained.

The top 100 scoring protein pairs were then evaluated for direct physical interactions using *AlphaFold3* (***Figure 2E***; ***Figure S9***; [37]). For each pair, we extracted the inter-protein *TM*-score (*ipTM*), a metric that quantifies the confidence and quality of the predicted protein complex structure—higher *ipTM* values indicate a greater likelihood of a true physical interaction.

To benchmark the *Bacformer* predictions, we also randomly sampled 100 neighbouring gene pairs (within five genes of each other) and 100 non-neighbouring gene pairs (separated by more than five genes). All sets—*Bacformer*-predicted and random—were evaluated with *AlphaFold3*, and their ipTM scores were extracted. The analysis of ipTM scores was stratified by gene proximity: pairs were classified as either "neighbours" (within five genes of each other) or "non-neighbours" (separated by more than five genes). Within each category, the ipTM scores of *Bacformer*-predicted pairs were compared to those of randomly selected pairs using Welch's *t*-test to determine whether *Bacformer*'s predictions were significantly enriched for high-confidence interactions (***Figure S9***).

The results demonstrate that *Bacformer* has strong predictive capacity for generating hypotheses about gene-gene interactions and gene networks, offering a valuable guide for experimental validation. This is particularly notable given that the model relies on *STRING DB*, which integrates diverse evidence sources and may include interactions lacking high-quality experimental support, leading to some false positives. *AlphaFold3*'s state-of-the-art computational predictions enable further screening of these predictions.

Uncovering gene networks from contextual embeddings. To test whether *Bacformer*'s contextualised representations predict gene co-expression and co-regulated genes, we compared zero-shot clusters obtained from pairwise cosine similarities between *Bacformer* embeddings with those from experimental co-expression data. Normalised RNA-seq data from 411 samples of P. aeruginosa were obtained from iModulonDB [38, 39] and used to compute a gene–gene co-expression matrix. Equivalent matrices were produced from *Bacformer* and from *ESM-2* embeddings to benchmark *Bacformer* results. Each matrix served as input to the Leiden clustering algorithm [23] across multiple resolutions. Cluster similarity between co-expression and model-predicted partitions was quantified using normalised mutual information (NMI, range 0–1; ***Figure S9***). For qualitative inspection, we extracted clusters at resolution 40 containing key regulators such as *lasR* (***Figure 2F***), *fleQ* (***Figure S9***), and *algU* (***Figure S9***).

Benchmarking. For Benchmarking, we evaluated the quality of the representations from each model on the protein-protein interaction task. We randomly sampled a set of 267 bacterial strains present in STRING DB [36], downloading the genomic assemblies from NCBI RefSeq [35] and matching the interaction labels from STRING DB using the locus tags present in the assemblies. We then randomly split the data into 60 / 20 / 20 train / validation / test split. We set the binary threshold at 0.6 using the Combined interaction score from STRING DB, so that all interactions >= 0.6 are treated as positive and all of the interactions <0.6 are negatives. We followed the approach described in the *Embedding bacterial genomes section* above to encode the gene/proteins.

For the **zero-shot** setting, given the embedding of two proteins, we compute the cosine similarity between the two and use the binary labels from STRING DB to compute the metrics. We report the results on the test set.

In the **finetuning** setting, we trained a linear classifier on binary labels (1 = interaction, 0 = no interaction) to predict whether the two proteins interact. Each gene/protein is represented by its frozen embedding. The pair of proteins are averaged and processed by a single-layer neural network preceded by layer normalization and dropout (rate 0.2). In addition to the average of the protein embeddings, we also include in the input the cosine similarity between the two embeddings, which we have seen to empirically improve performance on the validation set, therefore the input to the linear layer is $D + 1$, where D is the model dimensionality and 1 is the cosine similarity value. The

model is optimized with binary cross-entropy loss. We ran a learning rate sweep for each model on the validation set. Training proceeds for up to 20 epochs with early-stopping patience of 5 based on AUROC. We use Adam [14] optimizer with weight decay 0.01. We report results on the test set.

The datasets used for benchmarking are available at https://huggingface.co/datasets/macwiatrak/bacbench-ppi-stringdb-dna-small (DNA) and https://huggingface.co/datasets/macwiatrak/bacbench-ppi-stringdb-protein-sequences-small (protein sequences). The code for zero-shot and fine-tuning evaluation is available at https://github.com/macwiatrak/BacBench/tree/main/bacbench/tasks/ppi. Full table with results is available in ***Table S10***.

Protein function annotation

Gene essentiality prediction and benchmarking. Gene-essentiality annotations were downloaded from the *Database of Essential Genes* (*DEG*, http://origin.tubic.org/deg) [40]. Corresponding genomes (DNA and protein) were retrieved from NCBI GenBank. Of the 66 *DEG* genomes, those with > 98% redundant annotations were removed, leaving 51 genomes across 37 species. Data were randomly split into training, validation and test sets in a 60/20/20% ratio. The preprocessed gene essentiality dataset is available at https://huggingface.co/collections/macwiatrak/bacbench.

For each model we trained a linear layer preceded by layer normalisation and dropout of 0.2. The model was trained with binary cross-entropy loss to predict gene essentiality (1 = essential, 0 = non-essential) from the gene/protein embeddings; encoder weights were frozen. Training lasted up to 100 epochs with early-stopping patience 10, monitoring macro-AUROC. Adam weight decay was set to 0.01. For Evo embeddings we compared (i) log-likelihood vectors $z \in \mathbb{R}^N$ produced by the official scoring script and (ii) averaged hidden-state vectors $x \in \mathbb{R}^D$ ($D = 4,096$) obtained by modifying the *Evo* code. Option (i) performed better on the validation set and was used in final benchmarks. Full table with results is available in ***Table S11***.

AUROC and AUPRC curves per species were generated by selecting five strains from the test set and plotting predicted versus true labels per gene using the trained *Bacformer*(***Figure S10***). To assess the contribution of context alone, we repeated the experiment after masking all essential genes and 20% of non-essential genes, training the same linear layer only on the remaining masked embeddings and comparing results to the unmasked baseline (***Figure S10***).

To evaluate non-essential genes with the highest predicted probability of essentiality, we extracted the binary predictions on two strains from the test set—(1) *Staphylococcus aureus* (NC_002952) and (2) *Salmonella enterica* (NC_003197)—using finetuned *Bacformer* models trained on three different random seeds. The predictions for the two strains are available in ***Tables S1*** & ***S2***.

Protein function prediction. To evaluate *Bacformer's* capability to predict protein function and benchmark against other methods, we randomly sampled a set of 300 bacterial strains from KEGG (https://www.kegg.jp/) [41] and retrieved the annotations for each genome, together with the associated genome assemblies from NCBI RefSeq [35]. We then matched the gene/protein annotations from KEGG with the genome assemblies. This resulted in 590,657 annotated proteins out of a total of 1,126,510 (many present in each genome are annotated) 9,059 unique KEGG Orthology (KO) groups. The task was a multi-class classification problem with 9,059 classes. The 300 annotated strains were then split randomly into 60 / 20 / 20 train, validation and test sets respectively. We chose the lowest level of annotations in KEGG, the KO groups as our labels. We report macro metrics to ensure the models perform across distinct KO groups.

To evaluate **zero-shot** performance of different methods, we used a k-nearest neighbor classifier (kNN) using the gene/protein embeddings extracted from each model. We embedded all of the genes present in the dataset for each model, and evaluated the performance for different values of k (3, 5, 7), fitting the kNN model on the training dataset and evaluating on the validation set. We then used the k with the highest macro F1 score to select the best k. Finally, we evaluated the zero-shot performance on the test set. We used the NVIDIA cuML library [9] to run accelerated kNN algorithm.

For **fine-tuning**, we stacked a linear layer on top of the frozen gene/protein embeddings, preceded by a layer normalisation and dropout of 0.2 and trained it to predict the protein function label (KO group). We trained the model for 40 epochs and early stopping patience of 4 monitoring macro F1 score on the validation set. We performed a learning rate sweep for each model on the validation set. We employed Adam optimizer with weight decay of 0.01 and cross-entropy loss for model training. Full table with results is available in ***Table S12***.

Annotating proteins with unknown function. A large number of proteins across diverse bacterial genomes lack functional annotation, including many labelled as hypothetical. We fine-tuned the entire *Bacformer-L* (encoder + classification head) model and achieved >98% macro F1 and macro accuracy across strains on the held-out test set. We then used the fine-tuned model to infer labels for proteins without KEGG annotations. For each protein, we took the top 1 class from the softmax over more than 9,000 classes and required that its predicted probability be at least 0.20. With thousands of classes, this is a conservative cutoff: random chance for any given class is on the order of $\frac{1}{7{,}000}$, so a 0.20 posterior reflects strong model confidence and prioritizes precision while leaving uncertain cases unassigned. We assigned the predicted class to all proteins whose top probability exceeded this threshold.

To visualize the results, we embedded protein vectors from the fine-tuned *Bacformer-L* using UMAP and colored by (i) true label for annotated proteins and (ii) annotated vs unannotated status. We plotted the UMAP using Scanpy package using 15 neighbors for the kNN graph and minimum

distance of 0.5 for the UMAP.

The dataset of 300 strains for protein function prediction are available at https://huggingface.co/datasets/macwiatrak/bacbench-protein-function-kegg-dna-small (DNA) and https://huggingface.co/datasets/macwiatrak/bacbench-protein-function-kegg-protein-sequences-small (protein sequences). The code for benchmarking is available at https://github.com/macwiatrak/BacBench/protein_function.

Phenotypic traits prediction

Unsupervised clustering of bacterial genomes by phenotype. To evaluate whether bacterial genomes cluster by phenotype without supervision, we considered two traits: (1) environmental niche and (2) gram-stain. For environmental niche, we focused on the *Bifidobacterium* genus, chosen because its members colonise a range of mammalian gastrointestinal tracts—spanning infant, adult and ruminant guts—and must adapt to the distinct nutrient profiles and immune pressures found in each host. Strains were extracted from the *MGnify* data set [11] and 70 were randomly sampled across three environments. For every strain we computed a *Bacformer*genome embedding (model pre-trained on MAGs with a masked objective) and calculated pairwise cosine similarity between all strains.

For gram-stain, we used the annotations from [42] to cluster strains by gram-stain. After filtering for genomes with a gram-stain label (see below), 14,052 genomes remained. Each genome was embedded with *Bacformer*pre-trained on complete genomes, the genome-level vector being the mean of all protein embeddings. The embeddings were then projected with UMAP and the two-dimensional plot was coloured by gram-stain.

Phenotypic traits prediction and benchmarking. We collated large, public trait inventories into a unified benchmark, drawing on Madin et al. [42], Brbić et al. [43], and Weimann et al. [44]. To make provenance explicit and avoid name collisions, each label is prefixed with its source. We restrict to categorical traits, discard traits with fewer than 500 labeled genomes, and remove classes with fewer than 50 samples. The resulting corpus comprises 139 discrete phenotypes across 24,462 genomes spanning 15,477 species. Genome DNA and protein sequences are retrieved from NCBI GenBank using taxonomy IDs and assembly accessions provided by the trait resources.

As often phylogentically similar bacteria share the same trait, we split by genus for each phenotype—placing all genomes from a genus into a single split—and evaluate on held-out genera. We use 70 / 10 / 20 train / validation / test partitions and average results over three independent runs with different random splits.

For every phenotype–method pair, we train a single-layer neural classifier with layer normalization and dropout 0.2 before the linear layer on top of the frozen genome embeddings from each model. Fine-tuning other models than *Bacformer* on whole-genome tasks is not feasible due to the computational

cost of using an entire genome as input (see Computational complexity analysis section). We used the cross-entropy loss and Adam optimizer with weight decay of 0.01. We set the maximum number of epochs to 100 and an early stopping patience of 10, monitoring the macro AUROC. We perform learning rate sweep for each model, choosing the best learning rate based on the validation set results. We report the macro results across classes averaged across the random seeds. Full table with results is available in *Table S13*.

The phenotypic traits dataset are available at https://huggingface.co/collections/macwiatrak/bacbench and code for benchmarking is available at https://github.com/macwiatrak/BacBench/tree/main/bacbench/tasks/phenotypic_traits.

Phenotypic traits annotation. To annotate MAG corpus, we trained a separate linear model for each phenotypic trait using *Bacformer* embedding as described above. As the phenotypic traits dataset is mainly trained on data from genomic isolates, not MAGs, we first examined potential distribution shifts, since the classifiers were trained on complete genomes whereas MAGs are fragmented and sometimes incomplete. *Bacformer* embeddings were generated for every MAG and each phenotype model was re-trained on the MAG-derived vectors, revealing only a minor decrease in balanced accuracy. Thus, the embedding space learned on MAGs generalised well to the original tasks. We then selected phenotypes whose labels could be transferred with reasonable confidence via strain lineage, annotated the entire MAG corpus and evaluated performance where ground-truth labels were available. Balanced accuracy remained above 94% across diverse phenotypes, indicating robustness to genome type.

Phenotypes were grouped by held-out balanced accuracy: five exceeded 93% (high confidence), thirteen lay between 80% and 93% (medium) and fourteen fell between 75% and 80% (low). Predictions for every confidence tier were generated for the full MAG collection and released with accompanying confidence flags. The prediction files are available for download at https://drive.google.com/drive/folders/1PSR0mcE2fTexP9ADDVgYz3lkQ_qcyEsy?usp=sharing.

Polymicrobial communities annotation. To annotate bacterial genomes by phenotypes in a single sample, we selected the metagenomic sample with the largest number of MAGs (*SRS1996016*; *MGnify* [11]; human gut), containing $5,811$ MAGs. We then annotated this sample with phenotypic-trait prediction following the approach above; however, instead of a linear head we used an XGBoost model trained on top of frozen *Bacformer* embeddings to speed up annotation across all traits. For each trained trait model, we produced continuous scores for every genome using: (i) the predicted probability of the positive class for binary traits; and (ii) the softmax score of the highest-scoring class for multi-class traits. Continuous outputs yielded more informative embeddings than discrete labels. The result was a MAG $\times$ trait matrix of shape $(5811, 139)$.

To examine whether annotated MAGs cluster by taxonomy, we kept MAGs from the 50 most prevalent species (using *MGnify* taxonomy) and built a *k*NN graph (k=10) on the continuous trait profiles, then computed UMAP (dist=0.5) and visualised points coloured by family. We also quantified which phenotypic traits drive the community structure by running PCA on the MAG $\times$ trait matrix and computing a variance-weighted loading score per trait for the top 20 principal components. This score ranks traits by their contribution to variance. Finally, we visualised Lactose utilisation which was one of the highest scoring phenotypes using the computed UMAP. We leveraged scanpy package [24] to compute the *k*NN, UMAP, PCA and for visualizations.

Identifying phenotype-specific genes. We traced phenotype-linked genes for four traits: sporulation, motility type, anaerobic growth and environmental niche. The first three used the complete-genome trait data described above; environmental-niche prediction required a new data set. Families, genera and species occurring in multiple habitats were chosen so that the model could not rely on coarse taxonomic signals. From this pool we selected *Lactobacillus johnsonii* species and the genus *Prevotella* for detailed study because they occupy distinct niches yet share close taxonomic ranks, facilitating separation of lineage effects from habitat-specific adaptations. The MAGs for both were extracted from *MGnify* [11].

A separate *Bacformer* model was finetuned for each phenotype. Each data set was randomly split into 60% training, 20% validation, and 20% test sets. For sporulation, motility, and anaerobic growth, we initialized from the *Bacformer* checkpoint pretrained on complete genomes; for environmental-niche models, we used the MAG-pretrained checkpoint. Input $\times$ Gradient attribution [45] was applied to produce an importance score for each gene in every test genome, by summing over the contextualized protein-embedding vectors. To assess biological plausibility, we averaged scores across all occurrences of each annotated protein, discarded genes lacking annotations, and compared mean scores of proteins matching relevant keywords (e.g., “flagellum,” “spore formation,” “anaerobe”) to the remainder.

Sporulation served as a convenient qualitative benchmark due to its well-characterized genetic machinery. Ranking proteins by attribution successfully recovered known sporulation factors—especially the *spo* gene set—among the highest-scoring hits. After excluding proteins lacking locus tags in at least three strains, we averaged the top 100 ranked proteins across strains, clustered them using the Leiden algorithm (resolution 0.1, $k = 5$), and identified four functionally coherent modules corresponding to successive stages of spore formation.

We repeated this procedure to investigate environment-specific adaptation in *L. johnsonii*. For each host gut, the 100 most important proteins were averaged across strains and visualized with UMAP, with points colored by environment. Manual inspection highlighted several proteins previously implicated in host-specific colonization, confirming that the model identified meaningful determinants of niche preference.

Because many genomes lack reliable gene names or product descriptions due to fragmentation and annotation errors, we repeated the analysis using the protein-family (i.e. clusters) labels assigned during pretraining. As all proteins carry such a label, this approach ensured no sequences were discarded. Within the *Prevotella* test set, we computed mean importance for each family, averaged the top 100 families per environment, and visualized the results with UMAP, again revealing clear environment-specific clusters.

Predicting anti-microbial resistance

We curated antibiotic susceptibility readings from the NCBI Antibiotic Susceptibility Test (AST) Browser (https://www.ncbi.nlm.nih.gov/pathogens/ast), which aggregates hundreds of thousands of AST records across a broad range of drugs and 26,052 isolates. We then derived two label types: *(i)* binary and *(ii)* regression for antibiotics that had more than 500 unique labelled genomes and more than 50 genomes in either class (sensitive S / resistant R). The final set spans 37 distinct antibiotics with binary labels and 54 for regression (i.e. MIC prediction). The nr of genomes for each drug, including number of susceptible and resistant strains is available in ***Table S14***.

Benchmarking. To compare the predictive power of various models' representations, we extracted the frozen genome representation for all proteins in the genome and averaged across the protein embeddings to create genome embeddings for each model. We trained a linear layer on top using 70/10/20 train/validation/test partitions and average results over three independent runs with different random splits.

For every antibiotic (i.e. drug)–method (binary / regression) pair and all genomes, we train a single-layer neural classifier with layer normalization and dropout 0.2 before the linear layer on top of the frozen genome embeddings from each model. We used Adam optimizer with weight decay of 0.01. We set the maximum number of epochs to 100 and an early stopping patience of 10, monitoring the validation AUROC for the binary setting and validation *R squared* for the regression case. We leverage binary cross-entropy and mean squared error losses respectively for training binary and regression antibiotic resistance prediction models. We perform learning rate sweep for each model, choosing the best learning rate based on the validation set results. We report the results averaged across the random seeds. Full table with results is available in Table S15.

The AMR datasets are available at https://huggingface.co/collections/macwiatrak/bacbench and code for benchmarking is available at https://github.com/macwiatrak/BacBench/tree/main/bacbench/tasks/antibiotic_resistance.

Identifying AMR-associated genes. *Bacformer* is the first genomic LM which enables deep leaning embedding representations of entire genomes for which fine-tuning on multiple individual

tasks (such as AMR prediction) is computationally feasible (see Computational complexity analysis section).

Fine-tuning *Bacformer* to predict AMR was performed for three example bacteria / antibiotic pairs, chosen to span a range of drug actions and phylogeny diversity and because each had more than 500 publicly available genomes in NCBI with at least 200 in both the S/R classes. These include *E. coli* resistant to ampicillin ($N = 5,876$ genomes with ASTs: $3,305$ S / $2,571$ R); *Acinetobacter baumannii* resistant to gentamicin ($N = 970$: 217 S / 753 R); *Salmonella enterica* resistant to tetracycline (N = $8,828$: $4,339$ S / $4,489$ R).

The genomes from each species were split into training / validation / test (60/20/20). We then fine-tuned the *Bacformer* transformer to predict AMR from average protein embeddings using a single-layer neural classifier with a layer for normalization, dropout 0.2, learning rate 0.00015, 10% warm up and Adam optimizer [14].

The end-to-end model was fine-tuned for up to $100,000$ steps with validation every 250 steps monitored by AUROC and early stopping after 30 validation steps. Finetuning on one A100 GPU took 18-24 hrs for each species / antibiotic pair.

Attribution scores for each protein were derived by in-silico knockout for each finetuned model (each species / drug pair) using Captum (v0.8 `captum.ai/api/feature_ablation.html`) [46] with masking of all dimensions of each protein embedding providing a summed score for each protein, one at a time (the attributions from all 480 ablated dimensions were summed and then divided by 480 to give an attribution –1:1 for each protein). This gives a score reflecting the logit for model prediction with the protein present, minus the prediction when the protein is absent (ablated).

AMR Finder Plus [47] was used to annotate the proteins in each genome and stored alongside their attribution scores. We used a one-sided *Mann-Whitney* U rank-sum test to determine if the attribution scores for each annotation group (i.e., each element symbol or gene name) were greater than the background distribution of all protein attributions (a one-sided statistical test). Given the large size of our dataset (over 1 million proteins per species), the sampling distribution of the *Mann-Whitney* U statistic is well-approximated by a normal distribution. We therefore calculated p-values based on z-scores of the U-statistics. The resulting *p*-values were adjusted for multiple comparisons using the *Benjamini-Hochberg* procedure to control the false discovery rate at 0.05 (*Tables S4* (E. coli and ampicillin), *S5* (A. baumannii and gentamicin), and *S6* (Salmonella and tetracycline)) for each species/drug pair.

De novo **genome generation**

Using a pretrained *Bacformer* model with autoregressive objective on complete genomes, we have computed perplexity, which is a metric that indicates how well the model predicts the next element in a sequence. We then analysed the performance across test genomes, including species not seen during training.

Generating genomes with broad range of protein functions. We assessed whether *Bacformer* can generate genomes that encompass a broad range of essential functions—an important advance over existing generative models that are limited to designing individual molecules [48, 49, 50, 51, 52] or relatively simple multi-component systems [53].

In its original formulation *Bacformer* represents each protein with a continuous embedding produced by a protein language model from the raw amino-acid sequence. This fine-grained representation allows residue-level reasoning, yet it renders generation impractical: a decoder (the autoregressive component that generates sequences in transformers)„ which takes a vector in $\mathbb{R}^D$ as input (where D is the dimension of the embedding), would also have to predict an exact vector output in $\mathbb{R}^D$, an unbounded task, rather than producing a symbol from a finite vocabulary.

To enable genome generation we discretise the input. A vocabulary $\mathcal{F} = \{1, \ldots, F\}$ of $F = 50,000$ discrete protein clusters is obtained by leveraging the results of unsupervised clustering used for pretraining. Each protein is mapped to its cluster index so that a genome f becomes a sequence

$$\mathbf{f} = (f_1, f_2, \ldots, f_N), \qquad f_i \in \mathcal{F}.$$

The model now looks up a learnable token vector $\mathbf{x}_i = E_{f_i}$ from an embedding matrix $E \in \mathbb{R}^{F \times D}$, exactly as in word-piece language models [13]. All other components of the architecture remain unchanged.

We initialise E with the rows of the protein-cluster *classification head* $W \in \mathbb{R}^{F \times D}$ learned during the autoregressive MAG pretraining. As W already captures the distinguishing features of each cluster in the same latent space, this transfer (i) aligns the new discrete-token model with the previously learned semantic manifold, (ii) speeds up convergence, and (iii) avoids catastrophic forgetting of cluster-specific information.

Starting from this initialisation we further pretrained the discrete-token model on the complete-genome corpus, following the identical optimisation schedule used for the continuous-embedding *Bacformer*. This procedure leverages uninterrupted chromosomal context while retaining compatibility with sequence generation tasks.

We prompted pretrained autoregressive protein cluster *Bacformer* to generate a set of protein sequences ordered by their chromosomal positions from a prompt of 500 proteins starting from the origin of replication for *Staphylococcus aureus* (NCBI RefSeq ID `GCF_001019415.2`) and *Escherichia coli* (NCBI RefSeq ID `GCF_003627195.1`). The origin of replication represents the natural starting point for bacterial chromosome replication, providing a biologically meaningful context for genome generation

We assessed functional diversity by mapping protein families to *KEGG* BRITE ontology categories [41]. We extracted *KEGG* annotations for bacterial genes by downloading gene annotations and

genome assembly links from NCBI RefSeq and GenBank via the *KEGG* website. We processed the downloaded genomes and embedded them using the *ESM-2* model (12 layers, 35M parameters) [2]. After discarding proteins without *KEGG* annotations, we obtained over 600,000 annotated proteins. We assigned *KEGG BRITE* functions to our protein clusters using k-nearest neighbor classification (k=7) based on centroid embeddings with Euclidean distance. We validated the annotation quality by examining a random sample of protein clusters and their k-nearest neighbor assignments, and verified that the overall distribution of *KEGG* protein functions across protein families resembled real-world distributions. For generated genomes (excluding the 500-protein prompt), we assigned each protein family a *KEGG BRITE* function and compared the distribution of true versus predicted *KEGG* labels. This analysis confirmed that *Bacformer* generates genomes with functional repertoires that closely match those of true genomes (***Figure 4B***; ***Figure S13***).

To ensure that the model can generate diverse genomes and does not simply recapitulate genomes seen during training set, we generated multiple genomes using the same prompt while varying the temperature parameter to introduce controlled diversity, and we computed the distance between the generated genomes and those in the pretraining corpus. The results (***Figure S13***) show that *Bacformer* generates diverse genomes that are not merely replicas of any single training genome.

Genome completion. To test whether *Bacformer* ability to recognise genomes with missing sections, we used the complete genome held-out set ($N = 9,688$) and systematically removed 5%, 10%, 25% or 50% of the proteins by trimming sequentially from the genome's end. Both truncated and intact genomes were embedded with *Bacformer* (trained on complete genomes that were not part of the held-out set, using the autoregressive objective) and placed in the same embedding space for comparison. For each truncated genome we measured how highly its intact counterpart ranked among all genomes in the embedding space, reporting the mean reciprocal rank (MRR), where 1.0 indicates perfect match.

To visualize the effect of missing data, we projected *Bacformer* embeddings of full genomes and their 50%-ablated versions onto two dimensions using UMAP (10 neighbors). As a non-contextual baseline, we performed the same procedure with averaged *ESM-2* protein embeddings, omitting the *Bacformer* encoder (***Figure S14***).

Retrieving protein sequences from contextual embeddings. *Bacformer* generates contextual protein representations by associating each protein with one of 50,000 pre-defined protein clusters, rather than encoding each unique protein sequence individually. This clustering approach addresses the challenge posed by the immense diversity of protein sequences, enabling the model to generalize across related proteins while maintaining computational tractability.

By inverse mapping each embedding to its corresponding protein cluster, it becomes possible to retrieve the native protein sequence—or a close homolog—from a large candidate pool. This is

particularly valuable for applications such as genome completion and synthetic biology, where identifying the most likely protein sequence from an embedding is essential.

To facilitate this mapping, *Bacformer*'s output embedding for each gene $\mathbf{h}_i \in \mathbb{R}^d$ is trained using a contrastive loss. This loss encourages the *Bacformer* embedding to closely match the *ESM-2* embedding of the same protein within a batch, while remaining distinct from embeddings of other proteins. As a result, the learned embedding space supports accurate retrieval of protein cluster identities from contextual representations.

To achieve this, both sets of embeddings are ℓ_2-normalised and similarity is measured across protein embeddings using the cosine similarity, where $\mathrm{sim}(\mathbf{u}, \mathbf{v}) = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\| \, \|\mathbf{v}\|}$ and temperature $\tau = 1$, the noise-contrastive loss for a positive pair $(\mathbf{h}_i, \mathbf{e}_i)$ is

$$\ell_i^{\mathrm{ctr}} = -\log \frac{\exp\big(\mathrm{sim}(\mathbf{h}_i, \mathbf{e}_i)/\tau\big)}{\sum_{j=1}^{N} \exp\big(\mathrm{sim}(\mathbf{h}_i, \mathbf{e}_j)/\tau\big)}, \qquad \mathcal{L}_{\mathrm{ctr}} = \frac{1}{N} \sum_{i=1}^{N} \ell_i^{\mathrm{ctr}}.$$

Masked-family prediction uses the usual cross-entropy with the true family labels at masked positions and the logits from *Bacformer*'s classification head. The combined training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{mask}} + \alpha \mathcal{L}_{\mathrm{ctr}}, \qquad \alpha = 0.5.$$

During fine-tuning we initialised from the MAG-pretrained *Bacformer* with masked objective, computed $\mathcal{L}_{\mathrm{mask}}$ on masked tokens, computed $\mathcal{L}_{\mathrm{ctr}}$ on all non-special tokens, and back-propagated the combined loss. At inference time we measured cosine similarity and recall@K between contextual embeddings and their ground-truth sequence embeddings, observing strong retrieval for unmasked proteins and a marked dependence on the transformer's ability to refine input vectors for masked positions.

We trained the model on complete genomes, initializing from a *Bacformer* pretrained on MAGs with the masked objective and following the same method for complete genome training. We computed $\mathcal{L}_{\mathrm{mask}}$ on masked tokens, $\mathcal{L}_{\mathrm{ctr}}$ on all proteins, and back-propagated the combined loss.

Using a trained model we ran inference on test genomes and calculated the cosine similarity between the real and predicted embeddings. For both masked and unmasked proteins, we measured the cosine similarity between the positive (true) and negative (false) protein embeddings. Furthermore, we assessed the model performance using the recall@K metric, which measures how often the correct embedding is found among the top K most similar embeddings. While the model performed strongly on unmasked proteins, it achieved significantly worse results in the task of retrieving the true embedding at low K values for masked proteins. This highlights the importance of the *Bacformer* transformer backbone's capacity to update protein embeddings, as the contextualized

representations differ significantly from the initial protein embeddings used as input.

Generating genomes with desired properties with conditional finetuning. To explore how *Bacformer* could be applied to generate entire bacterial genomes endowed with specific properties, we selected two categorical traits from, the phenotypic dataset described above: oxygen requirements (specifically *aerobic*, *anaerobic* and *facultative*); and optimal growth temperature (*thermophilic* (>80°C), *psychrophilic* (<20°C) and *mesophilic* (20-40°C)).

We randomly split each dataset into train, validation and test splits with 60/20/20% ratio split. For training we initialized the model weights with the protein cluster *Bacformer* model pretrained on complete genomes described in the *Generating genomes with broad range of protein functions* section above. For each phenotype, we introduced a property token at the start of each protein-family sequence (***Figure 4F***). The property token was intitalized randomly and trained from scratch during training. We then finetuned *Bacformer* to generate protein-family sequences conditional on a phenotype token monitoring the validation loss separately for each trait.

Following finetuning, we used each trained model to generate a representative subset of genomes ($N = 20$) from the test set given a prompt of a property token and a set of 500 protein families. For comparison, we did the same, however, without the property token. We then compared the generated protein families with and without the property token, excluding the 500 prompt tokens. To evaluate the difference between the protein family sequences generated with and without the property token, we associated the protein family tokens with *KEGG Orthology* (KO) groups [41]. Thus, allowing us to identify specific orthologous protein groups which differ between the generated sequences. To do this, we used the annotated proteins from *KEGG* for bacterial genomes (see *Generating genomes with broad range of protein functions*). As a protein may belong to multiple orthologous groups, we assigned protein cluster tokens, using its centroid, to an orthologous group if 2 out of 10 nearest neighbors from the space of annotated proteins had the group label. We used euclidean distance to compute the kNN. The threshold was selected after experimenting with different values and evaluating the distribution of the number of KO groups assigned to protein family tokens. After assigning each protein family token its KO groups, we aggregated the KO groups based on the generated protein families for the genomes generated with and without the property token and computed the $log2$ fold-change to evaluate the difference between the two. Finally, to evaluate the statistical significance, we performed Fisher's exact test [26] using the aggregated KO counts from the genomes with and without the property token. We then qualitatively analysed the positively and negatively enriched statistically significant KO groups ($p < 0.075$).

We would like to highlight that full validation of synthetic genome viability remains outside the scope of this study, requiring complex genome engineering [54], these findings underscore the promise of contextual, evolution-informed models for rational genome design, laying the groundwork for more sophisticated generative designs in microbial genomics.

SUPPLEMENTARY FIGURES

SUPPLEMENTARY TABLES

Table S1. (separate file) Gene essentiality prediction results for *Staphylococcus aureus* (NC_002952) from *Bacformer* model across random seeds.

Table S2. (separate file) Gene essentiality prediction results for *Salmonella enterica* (NC_003197) from *Bacformer* model across random seeds.

Table S3. (separate file) Environment-specific genes for *Lactobacillus johnsonii* across human, pig and chicken gut based on gradient saliency using *Bacformer* model.

Tables S4-S6. (separate files) Attribution Scores for AMR Finder Plus elements for *E. coli* resistant to *ampicillin* (Table S4), *A. baumannii* resistant to *gentamicin* (Table S5) and *Salmonella enterica* resistant to *tetracycline* (Table S6). Each table has rows for all AMR Finder Plus elements (including AMR and non-AMR related subtypes and AMR classes causal to the antibiotic and other AMR classes). The rows are ordered by the Rank Biserial (a measure of the rankings of attributions of genes in the group compared to all genes). The rate of resistance is defined as the percentage of genomes with the gene that are resistant to the drug. Element penetrance is defined by the percentage of genomes that have at least one copy of the gene (number of genomes with gene / total number of genomes). We note that the number of proteins per symbol may be higher than the number of genomes, reflecting multiple (duplicate or similar) genes with the same element symbol in the genome.

Model	Input	Variant / Checkpoint	Objective	Tokenisation	Params	dim	Training corpus	Max ctx.
Mistral-DNA [55]	DNA	`Mistral-DNA-v1-138M-bacteria`	Autoregressive	Byte-pair	138M	768	Bacteria	512
DNABERT-2 [56]	DNA	`DNABERT-2-117M`	Masked	Byte-pair	117M	768	Multi-kingdom	512
Nucleotide Transformer [57]	DNA	`nucleotide-transformer-v2-250m-multi-species`	Masked	*k*-mer	250M	768	Multi-kingdom	2,048
ProkBERT [58]	DNA	`neuralbioinfo/prokbert-mini-long`	Masked	*k*-mer	27M	384	Bacteria	4,096
Evo [53]	DNA	`evo-1-8k-base (1.1_fix)`	Autoregressive	Single nucleotide	6.5B	4,096	Multi-kingdom	8,192
ESM-2 [2]	Single protein seq.	`esm2_t12_35M_UR50D`	Masked	Single amino acid	35M	480	Multi-kingdom	1,024
ESM-C [3]	Single protein seq.	`esmc_300m`	Masked	Single amino acid	300M	960	Multi-kingdom	1,024
ProtBERT [21]	Single protein seq.	`prot_bert`	Masked	Single amino acid	420M	1,024	Multi-kingdom	1,024
gLM2 [22]	Mixed modality (DNA & protein seq.)	`tattabio/gLM2_650M`	Masked	Single nucleotide/amino acid	650M	1,280	Bacteria	4,096
Bacformer	Multiple protein seq.	`macwiatrak/bacformer-masked-complete-genomes*`	Masked	Single protein	26M	480	Bacteria	6,000
Bacformer-L	Multiple protein seq.	`macwiatrak/bacformer-large-masked-complete-genomes*`	Masked	Single protein	332M	960	Bacteria	6,000

Table S7. Summary of benchmarked models. "Max ctx." = maximum context length supported at inference; for DNA models measured in base pairs, for pLMs in amino acids, and for Bacformer in number of proteins present in the genome. "dim" = dimensionsionality of the output of the last hidden layer. *We use either `bacformer-masked-complete-genomes` or `bacformer-masked-MAG` depending on the genome type used as input.

Model	Max ctx.	DNA-seq overlap	Promoter length
Mistral-DNA	512	16	128
DNABERT-2	512	16	128
Nucleotide Transformer	2,048	32	128
ProkBERT	4,096	64	128
Evo	8,192	32	128
ProtBERT	1,024	N/A	N/A
ESM-2	1,024	N/A	N/A
ESM-C	1,024	N/A	N/A
gLM2	4,096	64	128
Bacformer	1,024 / 6,000*	N/A	N/A
Bacformer-L	1,024 / 6,000*	N/A	N/A

Table S8. Model-specific context parameters used in this study. "Max ctx." is the maximum input length; "DNA-seq overlap" is the stride between consecutive windows when sliding across a genome; "Promoter length" is the upstream sequence length concatenated for promoter prediction.* For Bacformer the maximum input size of a protein is $1,024$ amino acids and maximum number of proteins in the genome is set to $6,000$.

Model	ARI	NMI	ASW
gLM2	0.710 ± 0.028	0.836 ± 0.009	0.381 ± 0.012
ProtBERT	0.720 ± 0.028	0.866 ± 0.012	0.326 ± 0.037
ESM-2	0.520 ± 0.028	0.732 ± 0.007	0.191 ± 0.022
ESM-C	0.707 ± 0.022	0.861 ± 0.009	0.277 ± 0.031
Bacformer	0.857 ± 0.035	0.924 ± 0.015	0.250 ± 0.022
Bacformer Large	$\mathbf{0.970 \pm 0.017}$	$\mathbf{0.985 \pm 0.006}$	$\mathbf{0.460 \pm 0.031}$

Table S9. Clustering performance across species-level genome clustering, reported as Adjusted Rand Index (ARI), Normalised Mutual Information (NMI), and Average Silhouette Width (ASW). Best value for each metric is bolded. Mean $\pm$ standard deviation, where standard deviation was computed by performing 10 independent bootstrap resamples, each using 80% of the data.

Method	Zero-shot		Fine-tuned	
	AUROC	AUPRC	AUROC	AUPRC
Mistral-DNA	51.03 ± 2.88	14.11 ± 9.35	55.03 ± 2.35	16.17 ± 10.79
Evo	52.39 ± 2.06	14.18 ± 9.85	54.05 ± 2.18	15.91 ± 10.56
DNABERT-2	52.68 ± 4.62	15.59 ± 10.83	61.35 ± 3.08	24.60 ± 12.91
Nucleotide Transformer	56.20 ± 3.10	17.89 ± 12.50	61.55 ± 3.02	21.91 ± 12.66
ProkBERT	55.65 ± 5.56	17.49 ± 12.11	67.04 ± 2.59	31.36 ± 13.25
gLM2	54.11 ± 3.62	15.59 ± 10.61	71.99 ± 2.67	40.22 ± 12.91
ProtBERT	49.79 ± 4.01	13.62 ± 9.70	68.51 ± 3.91	35.13 ± 12.24
ESM-2	56.85 ± 1.88	19.51 ± 11.09	67.84 ± 3.64	32.05 ± 12.34
ESM-C	57.82 ± 3.27	18.13 ± 10.98	70.51 ± 3.64	37.04 ± 12.34
Bacformer	66.36 ± 4.64	28.74 ± 14.56	73.13 ± 2.40	40.16 ± 13.17
Bacformer Large	**72.08 ± 6.37**	**35.49 ± 17.45**	**77.64 ± 2.54**	**45.34 ± 14.02**

Table S10. Protein-protein interaction performance on the held-out test set (higher is better), median ± standard deviation across strains. Best value for each metric is bolded.

Method	AUROC	AUPRC
Mistral-DNA (138M)	55.80 ± 4.14	22.62 ± 7.82
Evo (6.5B)	68.15 ± 10.27	26.74 ± 14.25
DNABERT-2 (117M)	66.07 ± 8.32	29.81 ± 12.03
Nucleotide Transformer (235M)	66.83 ± 6.92	28.32 ± 9.40
ProkBERT (26M)	72.15 ± 10.33	37.94 ± 14.96
gLM2 (650M)	81.94 ± 9.44	**59.31 ± 13.13**
ProtBERT (412M)	77.79 ± 8.29	47.60 ± 12.00
ESM-2 (35M)	77.52 ± 8.15	38.07 ± 11.09
ESM-C (332M)	80.80 ± 9.15	53.15 ± 13.57
Bacformer (26M)	78.33 ± 7.61	49.41 ± 12.05
Bacformer (332M)	**82.04 ± 8.12**	53.53 ± 11.73

Table S11. Gene essentiality prediction performance on the held-out test set (higher is better), median ± standard deviation across strains. Best value for each metric is bolded.

Method	Zero-shot		Fine-tuned	
	Macro F1	Macro accuracy	Macro F1	Macro accuracy
Mistral-DNA	0.10 ± 7.90	0.27 ± 9.07	0.70 ± 5.35	1.38 ± 6.12
Evo	1.23 ± 3.66	2.03 ± 5.03	0.80 ± 4.25	1.20 ± 5.75
DNABERT-2	2.50 ± 11.66	3.78 ± 13.04	11.45 ± 14.17	16.40 ± 15.00
Nucleotide Transformer	6.06 ± 11.70	8.82 ± 13.22	19.73 ± 12.33	26.89 ± 12.93
ProkBERT	11.78 ± 14.12	15.71 ± 15.44	55.23 ± 16.32	64.45 ± 15.36
gLM2	**90.06** ± **12.04**	**93.66** ± **9.90**	97.01 ± 3.38	98.06 ± 2.21
ProtBERT	78.26 ± 14.47	84.65 ± 12.65	97.24 ± 2.95	98.31 ± 1.80
ESM-2	87.85 ± 9.88	92.04 ± 7.58	93.48 ± 5.04	96.06 ± 3.31
ESM-C	87.58 ± 16.60	91.34 ± 14.52	**97.35** ± **2.92**	**98.41** ± **1.96**
Bacformer	87.54 ± 9.34	92.04 ± 7.15	91.92 ± 6.39	94.82 ± 4.52
Bacformer Large	84.55 ± 16.40	89.59 ± 14.22	91.94 ± 10.34	94.50 ± 8.60

Table S12. Protein function prediction performance on the held-out test set (higher is better), median ± standard deviation across strains. Best value for each metric is bolded.

Method	AUROC	AUPRC	F1
Mistral-DNA	60.34 ± 9.81	39.72 ± 33.09	47.25 ± 6.67
Evo	60.11 ± 11.00	39.65 ± 33.27	46.52 ± 7.50
DNABERT-2	63.83 ± 11.06	42.10 ± 34.24	49.42 ± 8.99
Nucleotide Transformer	65.98 ± 11.45	43.22 ± 34.85	52.04 ± 11.07
ProkBERT	67.64 ± 11.87	44.80 ± 34.72	52.03 ± 11.56
ESM-2	68.71 ± 12.36	45.61 ± 35.45	53.10 ± 11.90
ESM-C	69.07 ± 11.98	45.54 ± 35.38	52.28 ± 11.39
ProtBERT	68.47 ± 11.97	45.47 ± 34.89	53.71 ± 11.65
gLM2	65.82 ± 11.41	44.00 ± 34.40	51.11 ± 10.86
Bacformer	71.34 ± 12.91	47.80 ± 35.99	56.14 ± 13.19
Bacformer Large	**73.41** ± **12.10**	**48.58** ± **12.89**	**59.65** ± **12.89**

Table S13. Overall phenotypic traits prediction performance across all phenotypes. Values are mean ± standard deviation across 3 random seeds. The results were macro-averaged across classes for each phenotype. Best value for each metric is bolded.

Drug	# Resistant	# Susceptible	# Total
amikacin	7353	588	7941
ampicillin	10052	6211	16263
amoxicillin–clavulanic acid	12458	1683	14141
azithromycin	10497	2509	13006
cefazolin	1666	1897	3563
cefepime	3237	1188	4425
cefotaxime	566	446	1012
cefoxitin	12463	3902	16365
ceftazidime	2362	717	3079
ceftriaxone	13385	3469	16854
ciprofloxacin	17353	4280	21633
clindamycin	4327	337	4664
ertapenem	7708	373	8081
erythromycin	5605	1860	7465
fosfomycin	1186	396	1582
gentamicin	18242	2951	21193
imipenem	8485	409	8894
kanamycin	3768	410	4178
levofloxacin	3302	1106	4408
meropenem	11320	811	12131
nalidixic acid	1262	780	2042
nitrofurantoin	10535	5316	15851
oxacillin	6929	762	7691
piperacillin–tazobactam	3294	968	4262
rifampin	611	354	965
streptomycin	11965	2442	14407
sulfamethoxazole	3770	707	4477
sulfisoxazole	1569	323	1892
tetracycline	6936	3242	10178
tigecycline	662	357	1019
trimethoprim	2617	582	3199
ampicillin–sulbactam	1632	451	2083
aztreonam	2530	728	3258
ceftaroline	611	152	763
chloramphenicol	5345	1016	6361
colistin	572	290	862
daptomycin	492	131	623

Table S14. Number of resistant, susceptible, and total labelled genomes for every antibiotic in the binary prediction setting.

Method	Binary		Regression	
	AUPRC	AUROC	R^2	Pearson
Mistral-DNA	49.86 ± 22.73	76.05 ± 9.14	19.71 ± 17.37	40.95 ± 21.51
Evo	44.06 ± 23.28	68.22 ± 9.32	10.01 ± 10.36	29.18 ± 17.26
DNABERT-2	52.35 ± 23.55	79.88 ± 7.26	23.61 ± 18.33	45.89 ± 19.27
Nucleotide Transformer	59.70 ± 22.90	84.22 ± 6.47	27.74 ± 17.80	51.19 ± 16.76
ProkBERT	58.90 ± 23.67	83.94 ± 6.39	28.00 ± 17.57	51.24 ± 16.91
ESM-2	62.04 ± 23.64	85.05 ± 6.63	31.18 ± 17.39	54.60 ± 16.15
ESM-C	63.41 ± 23.43	85.68 ± 6.81	33.15 ± 17.62	56.79 ± 15.46
ProtBERT	61.79 ± 23.19	84.51 ± 6.31	28.23 ± 18.46	51.56 ± 17.61
gLM2	55.80 ± 24.46	82.16 ± 6.75	26.15 ± 18.13	49.41 ± 17.14
Bacformer	67.97 ± 20.73	87.61 ± 5.68	33.84 ± 18.60	57.53 ± 15.43
Bacformer Large	**76.14 ± 19.59**	**92.14 ± 4.20**	**42.59 ± 19.54**	**65.73 ± 12.88**

Table S15. Performance (percent, higher is better) on the antibiotic resistance prediction tasks across drugs. Values are mean ± standard deviation across 3 random seeds; best value for each metric is bolded.

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